MultiPartyTurnBench: Benchmarking Floor Control in Multi-Party Spoken Dialogue

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Abstract

Full-duplex speech systems must decide when to enter, hold, or relinquish the conversational floor, even as multiple speakers overlap or backchannel without taking it. Existing evaluations study dyadic turn-taking or end-to-end agent behavior, but lack a unified, controlled evaluation of genuine floor transitions against confusable non-events in multi-party settings. We introduce **MultiPartyTurnBench** (MPTB), a benchmark of 240 synthetically generated, channel-separated conversations spanning two to five speakers and six interaction types. MPTB evaluates two complementary tasks: End-of-Turn (EOT) detection distinguishes genuine turn completion from temporary holds, while Interruption (INT) detection distinguishes successful floor takeovers from non-floor-taking speech. Exact event-conditioned annotations enable causal evaluation using false-positive-constrained recall and signed latency. Across acoustic, semantic, and turn-projection baselines, we find a stark asymmetry: learned systems achieve 82.5–92.6% EOT recall at 4.3–11.8% false-positive rates, while INT detection suffers from missed takeovers or excessive false activations. This asymmetry extends to five full-duplex agents which achieve 86.1–95.8% balanced accuracy on turn entry, yet perform near chance (46.9–54.7%) on selective interruption handling. Gemini-3.8-Live yields to 97.7% of backchannels, whereas MiniCPM-o-4.5 yields to only 1.4% of genuine interruptions. These findings establish selective floor control as a distinct, unresolved challenge for reliable multi-party voice interaction.

1 Introduction

Full-duplex speech systems must continuously decide when to enter, hold, or relinquish the conversational floor while processing ongoing speech (Ekstedt and Skantze, 2022; Peng et al., 2025). These decisions require more than detecting speech activity: a speaker may pause without yielding the floor, while a listener may backchannel without attempting to take it (Sacks et al., 1974; Yngve, 1970). Misinterpreting either can cause an agent to respond prematurely or unnecessarily abandon its own response, even when its language understanding is otherwise correct (Xia et al., 2026).

This distinction becomes particularly important in multi-party conversation, where several listeners may backchannel, overlap, or interject while the current speaker retains the floor. Existing evaluations address complementary aspects of this problem: dyadic benchmarks distinguish turn endings and interruptions in 2-party settings (Jiang et al., 2026; Xia et al., 2026), natural multi-party corpora study next-speaker prediction (Triadic-VAP, 2025; Qi and Skantze, 2026), and full-duplex agent benchmarks assess speaking and stopping behavior (Lin et al., 2025, 2026c; Ma et al., 2026). However, the joint evaluation of genuine floor relinquishment and successful floor acquisition under controlled multi-party conditions remains underexplored.

We introduce **MultiPartyTurnBench** (Fig. 1), a benchmark for evaluating floor control in multi-party full-duplex speech. MPTB evaluates two complementary decisions: End-of-Turn (EOT) detection distinguishes genuine turn completion from temporary holds, while Interruption (INT) detection distinguishes successful floor takeovers from backchannels and other non-floor-taking speech. The benchmark comprises 240 synthetically generated, channel-separated conversations balanced across $N \in \{2, 3, 4, 5\}$ speakers and six interaction types. A discrete-event orchestrator establishes floor ownership and event timing before speech synthesis, enabling controlled variation in speaker count, interaction dynamics, and acoustic conditions. We evaluate acoustic, semantic, and turn-projection systems through causal event detection with false-positive-constrained recall and signed

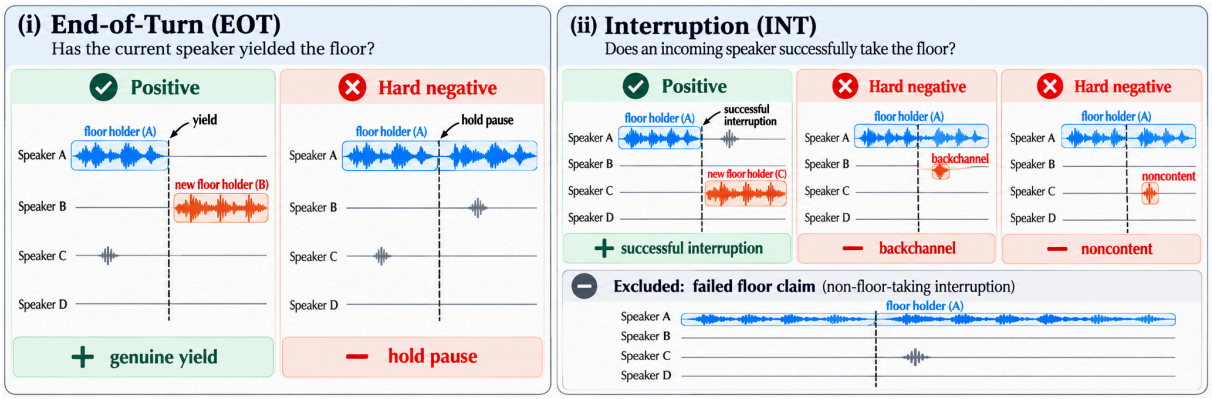


Figure 1: **MultiPartyTurnBench evaluation tracks.** The End-of-Turn (EOT) track distinguishes genuine floor relinquishment from mid-turn pauses where the speaker retains the floor. The Interruption (INT) track distinguishes successful floor takeovers from backchannels and non-content vocalizations that leave floor ownership unchanged. Non-floor-taking interruption attempts are excluded from causal INT scoring because their ambiguous outcomes become apparent only through subsequent interaction.

latency, complemented by real-speech validation and interactive testing of full-duplex agents.

Main Results. Our evaluation reveals a pronounced asymmetry between turn completion and successful floor acquisition. At qualified operating points, systems recover up to 92.6% of turn endings but only 56.2% of successful interruptions; acoustic detectors achieve higher recall only at excessive false-positive rates. Increasing speaker count produces jagged degradation, while conversational style exposes substantial model-specific interruption failures. Crucially, this asymmetry extends to observable voice agent behavior: four of five full-duplex agents achieve 86.1–95.8% balanced accuracy on turn entry, yet all five remain near chance (46.9–54.7%) on selective yielding. Gemini-3.8-Live (Google, 2026) yields almost indiscriminately, whereas MiniCPM-o-4.5 (Cui et al., 2026) rarely yields even to genuine interruptions, revealing opposite failures concealed by aggregate accuracy. Complementary real-speech evaluation further shows that interruption detection remains challenging in natural multi-party meetings. These findings establish selective interruption handling as a persistent limitation of multi-party full-duplex interaction.**Our main contributions are:**

- **MultiPartyTurnBench**, a controlled benchmark of 240 multi-party conversations spanning two to five speakers and six interaction types, with exact event-state annotations for end-of-turn (EOT) and interruption (INT) detection.
- **A systematic evaluation of floor-transition detection** across acoustic, semantic, and turn-projection systems, revealing an EOT–INT capability gap and model-specific sensitivities to speaker count and conversational style.
- **End-to-end validation across five full-duplex voice agents**, exposing near-chance selective yielding despite strong turn-entry performance, with opposing tendencies to yield indiscriminately or ignore genuine interruptions.

2 Related Work

Turn-taking and interruption benchmarks. Conversational floor management involves turn transitions, interruptions, backchannels, and competing speaker claims (Sacks et al., 1974; Duncan, 1972; Yngve, 1970). Talking Turns (Arora et al., 2025a) evaluates turn-taking in audio foundation models, while TurnBench (Jiang et al., 2026) established causal EOT and interruption detection in dyadic conversational styles. SID-Bench (Xia et al., 2026) further studies semantic interruption detection against non-interruptive backchannels. Table 1 shows that **`MultiPartyTurnBench` addresses the complementary multi-party setting**, where several listeners may produce overlapping, non-floor-taking speech and successful floor acquisition must be distinguished from competing vocal activity under a common event-level evaluation protocol.

Multi-party turn-taking resources. Natural multi-speaker corpora support complementary prediction tasks. TEIDAN (Triadic-VAP, 2025) evaluates future speaker activity in triadic Japanese conversations. AVCC (Qi and Skantze, 2026) evaluates shift/hold and next-speaker prediction in audiovisual dialogue. VoxConverse (Chung et al., 2020) studies floor-shift detection, boundary timing, and next-speaker identification (Fast When, Careful Who, 2026). These resources capture natu-

Benchmarks	Multi-party	EOT event detection	INT event detection	Event-specific negatives	Timestamped event scoring	FPR budget	Controlled Speaker Comparison	Agent behavior scoring
Turn-taking and interruption benchmarks								
TurnBench (Jiang et al., 2026)	✗	✓	✓	✓	✓	✓	✗	✗
Talking Turns (Arora et al., 2025a)	✗	✗	✗	✗	✗	✗	✗	✓
SID-Bench (Xia et al., 2026)	✗	✗	✓	✓	✓	✗	✗	✗
ECHO (Zhao et al., 2026b)	✗	✗	✗	✓	✗	✗	✗	✓
Multi-party turn-taking evaluation resources								
TEIDAN (Triadic-VAP) (Triadic-VAP, 2025)	✓	✗	✗	✗	✗	✗	✗	✗
AVCC (MuVAP) (Qi and Skantze, 2026)	✓	✗	✗	✗	✗	✗	✓	✗
Fast When, Careful Who (Fast When, Careful Who, 2026)	✓	✓	✗	✓	✓	✗	✓	✗
Full-duplex and multi-speaker agent benchmarks								
Full-Duplex-Bench (Lin et al., 2025, 2026c,b)	✗	✗	✗	✓	✗	✗	✗	✓
FD-Bench (Peng et al., 2025)	✗	✗	✗	✗	✗	✗	✗	✓
Duplex-MPE (Ma et al., 2026)	✓	✗	✗	✗	✗	✗	✗	✓
MultiTalkBench (Wang et al., 2026)	✓	✗	✗	✗	✗	✗	✗	✓
MSI-Bench (Xiong et al., 2026)	✓	✗	✗	✗	✗	✗	✗	✓
MultiPartyTurnBench (ours)	✓	✓	✓	✓	✓	✓	✓	✓

Table 1: **Comparison of turn-taking and full-duplex evaluation benchmarks.** Existing benchmarks address dyadic turn-taking events, natural multi-party next-speaker predictions, or agent interaction behavior. `MultiPartyTurnBench` jointly evaluates multi-party EOT and INT event detection with event-specific hard negatives, timestamped causal decisions, FPR-constrained scoring, controlled speaker-count comparisons, and complementary agent behavior validation.

Event	Floor outcome	EOT	INT
TURN: yield	Released	✓	–
hold	Retained	×	–
interrupted	Taken over	–	–
INTERRUPTION	Transferred	–	✓
NON-FLOOR-TAKING INT.	Retained	–	Excl.
BACKCHANNEL	Retained	–	×
NONCONTENT	Unchanged	–	×

Table 2: **Floor-state event taxonomy in `MultiPartyTurnBench`.** Events are mapped to EOT and INT evaluation targets: ✓ denotes a positive event, × a hard negative, and – an event not scored in that track. Excl. denotes events explicitly excluded from scoring.

ral conversational dynamics, but do not score end-of-turn and interruption detection.

Full-duplex agent benchmarks. Full-Duplex-Bench series (Lin et al., 2025, 2026c,b) and FD-Bench (Peng et al., 2025) evaluate real-time conversational behavior. Recent multi-party benchmarks broaden this scope in multi-party settings: Duplex-MPE (Ma et al., 2026) assesses if agents respond, remain silent, or stop speaking in group conversations; MultiTalkBench (Wang et al., 2026) examines long-form multi-party dialogue, and MSI-Bench (Xiong et al., 2026) evaluates speaker-grounded memory, instruction following, and reasoning. These benchmarks primarily assess conversational outcomes rather than standalone floor-event predictions. **MultiPartyTurnBench isolates the underlying floor-discrimination problem**, while separately validating how these distinctions manifest in full-duplex agent behavior.

3 `MultiPartyTurnBench`

`MultiPartyTurnBench` evaluates whether systems can causally infer changes in *conversational floor ownership* from ongoing multi-speaker audio. It defines two complementary, event-conditioned tasks: **(i) End-of-Turn (EOT)** asks whether the *current speaker has relinquished the floor* and aims to detect genuine floor relinquishment by the current speaker, anchored at speech offset against hard negatives such as mid-turn pauses where the speaker retains the floor. **(ii) Interruption (INT)** asks whether *another speaker has successfully acquired the floor* and detects successful floor acquisition by another speaker, anchored at the interrupting speech onset, against hard negatives like backchannels and non-content vocalizations that leave floor ownership unchanged. Unsuccessful interruption attempts are excluded because their outcomes may only become apparent through subsequent interaction, making them ambiguous at onset for a causal detector (Jiang et al., 2026). Together, the two tracks evaluate *floor relinquishment and acquisition* against behaviorally confusable non-events, rather than speech activity alone (Table 2).

3.1 Benchmark Construction

MPTB is designed to isolate floor-control decisions under independently controlled interaction conditions with exact event boundaries. In natural multi-party recordings, speaker count, overlap, listener activity, and acoustics vary jointly, complicating controlled analysis. We construct each conversation from an explicit floor-state representation before rendering speech in three steps: (1) define

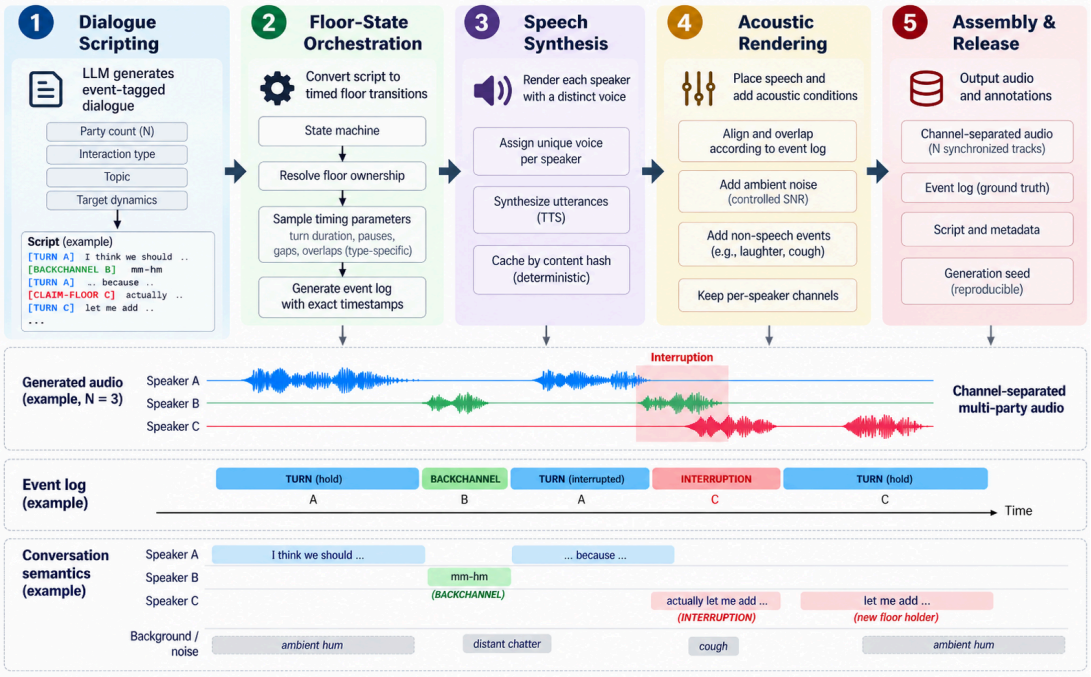


Figure 2: **MultiPartyTurnBench conversation generation pipeline.** Five-stage pipeline transforms event-conditioned dialogue scripts into synchronized multi-speaker audio with exact floor-transition annotations. A discrete-event orchestrator resolves floor ownership and temporal dynamics before speech synthesis, followed by speaker rendering, acoustic augmentation, and channel assembly. The example illustrates overlapping speech, backchannels, successful interruptions, and their event annotations.

the event taxonomy for floor ownership changes; (2) select the interaction conditions and dialogue scripts; (3) staged generation pipeline renders synchronized multi-speaker audio recordings.

Event Taxonomy. Each event is defined relative to the current floor holder and whether floor ownership is retained, relinquished, or transferred. Five event types capture these transitions: TURN represents speech by the floor holder, resolved as `hold`, `yield`, or `interrupted`; INTERRUPTION denotes a successful listener floor claim, while NON-FLOOR-TAKING INTERRUPTION denotes an unsuccessful attempt. BACKCHANNEL captures listener speech without claiming the floor, and NON-CONTENT captures non-linguistic activity without changing floor ownership. Table 2 maps these events to the EOT and INT evaluation targets.

Controlled Interaction Axes. MPTB crosses speaker count ($N \in \{2, 3, 4, 5\}$) with six interaction types derived from TurnBench (Jiang et al., 2026): Casual, Task-Oriented, Instructional, Collaborative, Argumentative, and Narrative. These regimes elicit distinct turn-taking dynamics while enabling controlled comparisons across group sizes at fixed interaction style, and vice versa. Generation additionally controls backchannel and interruption occurrences, while acoustic rendering varies ambient noise and nuisance events. See Appendix C.2 for definitions of each interaction type.

Conversation Generation Pipeline Figure 2 illustrates the five-stage pipeline for multi-speaker acoustic realization annotated with conversational floor dynamics. The first two stages establish the canonical event log and interaction timeline before speech synthesis, ensuring that ground-truth labels derive from the underlying floor state rather than retrospective waveform analysis.

1. Event-conditioned dialogue generation. Off-the-shelf LLM like GPT-4.1 generates multi-party scripts conditioned on speaker count, interaction type, topic, narrative plot, and target turn-taking statistics. A restricted event vocabulary (`TURN`, `CLAIM_FLOOR`, `VOCALIZE_NO_CLAIM`) specifies conversational actions, with backchannel and interruption frequencies guided by TurnBench (Jiang et al., 2026). The LLM determines linguistic content and proposes listener actions but does not assign final timestamps or resolve floor ownership. Scripts are schema-validated, with malformed outputs regenerated or rejected.

2. Floor-state orchestration. A discrete-event orchestrator maintains the current floor holder and resolves symbolic actions into realized transitions: `TURN` becomes a hold, yield,

or interruption-forced termination; CLAIM_FLOOR resolves either to a successful INTERRUPTION or a NON-FLOOR-TAKING INTERRUPTION; and VOCALIZE_NO_CLAIM produces a backchannel or related non-floor-changing event. The orchestrator assigns turn durations, inter-speaker gaps, pauses, and interruption overlap based on real-world timing statistics from AMI corpus (Carletta et al., 2006). Explicit mid-turn pauses provide EOT hard negatives. The resulting placement schedule and canonical event log define exact EOT and INT anchors.

3. Speaker rendering. Utterances are independently synthesized using Kokoro-82M(hexgrad, 2024), with distinct voices sampled from a gender-balanced pool of 24 voices, preventing voice reuse within a conversation. Development and test splits use disjoint voice inventories (8 and 16 voices, respectively). Word-level timing is retained for alignment and quality control.

4. Acoustic scene rendering. Synthesized speech is placed on synchronized speaker channels according to the orchestrated timeline, preserving overlap determined by floor-state transitions. Successful interruptions end with a short fade applied at the truncation boundary, and channels are peak-normalized independently. A shared voice-free ambient bed (filtered pink noise and humming) is mixed at a per-conversation SNR. MUSAN (Snyder et al., 2015) sound effects and intermittent distant third-party speech introduces nuisance activity to stress false-positive detection; neither creates benchmark-positive floor events.

5. Channel assembly and release artifacts. Each conversation is released with its noisy evaluation channels, clean reference channels, tagged script, canonical event log, render metadata, and generation seed. Clean channels are retained for quality control, while benchmark systems operate on the noisy rendition. All non-generative stages are deterministic with recorded seeds and manifests, enabling exact reproduction of the interaction schedule and acoustic assembly (see Appendix C.3).

3.2 Benchmark Composition

MPTB contains 240 conversations (6.89 hours), balanced across four speaker-count settings ($N \in \{2, 3, 4, 5\}$) crossed with the six interaction types. It contains 48 development and 192 test conversations using disjoint speaker voices. The corpus contains 3,422 EOT positives and 241 hold negatives, alongside 367 INT positives and 1,481 non-floor-taking negatives. MPTB closely mimic natural turn-taking distributions. See Appendix D for event-interaction trends. We additionally validate the generated floor events through independent human annotations in Appendix A).

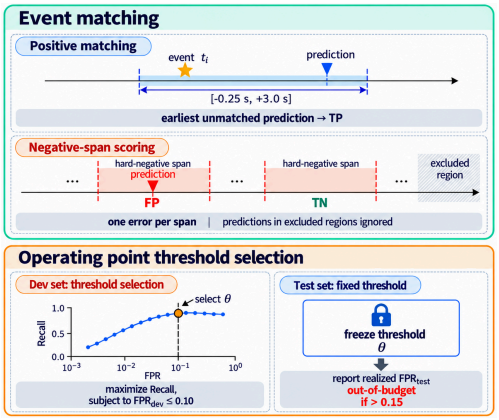


Figure 3: **`MultiPartyTurnBench` evaluation protocol.** Causal per-speaker predictions are matched to reference events and scored against hard-negative spans. Development-selected thresholds are frozen for test evaluation, with performance measured through recall, false-positive rate, and latency.

4 Evaluation

Figure 3 illustrates the causal evaluation protocol: systems emit timestamped EOT and INT decisions using only audio observed up to prediction time. We evaluate event recall, false-positive rate, and signed decision latency at fixed operating points.

Prediction and Event Matching. Per-speaker model scores are converted to discrete events through threshold crossings with a 2 s refractory period. Following TurnBench (Jiang et al., 2026), each gold positive at t_i is matched one-to-one to the earliest unmatched prediction within $[t_i - 0.25,\ t_i + 3.0]$ s. Matched events are true positives (TP), while unmatched gold positives are false negatives (FN). Hard negatives are scored as temporal spans: any detection within a span counts as a false positive (FP), otherwise a true negative (TN). Predictions outside scored regions are ignored.

Operating Points and Metrics. For each model, we select the development threshold maximizing recall subject to $\mathrm{FPR}_{\mathrm{dev}} \leq 0.10$ (Jiang et al., 2026), then freeze it across all test conversations, speaker counts, and interaction types. Systems exceeding $\mathrm{FPR}_{\mathrm{test}} > 0.15$ are marked out-of-budget. We report $\mathrm{Recall} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ and $\mathrm{FPR} = \mathrm{FP}/(\mathrm{FP} + \mathrm{TN})$, alongside signed latency $\Delta t_i = t_i^{\mathrm{pred}} - t_i^{\mathrm{gold}}$ (p10/p50/p90), where +ve / −ve values indicate delayed / early commitments.

Full-Duplex Behavioral Diagnostics. For full du-

plex voice agents, recall alone cannot distinguish indiscriminate yielding from selective floor control. We additionally report balanced accuracy $\mathrm{bACC} = (\mathrm{Recall} + 1 - \mathrm{FPR})/2$ and action bias $A_{\mathrm{bias}} = \mathrm{Recall} + \mathrm{FPR} - 1$. Balanced accuracy measures discrimination between genuine floor transitions and hard negatives. Action bias indicates excessive responsiveness (+ve) or reluctance to act (−ve). More details in Appendix B.2.

5 Experiments

Offline EOT and INT Evaluation. We evaluate 12 systems across three families: **acoustic VAD** (RMS Energy, WebRTC VAD (Google WebRTC Project), Silero VAD (Silero Team, 2024)); **semantic endpointers** (Smart Turn v2 (Pipecat AI, 2025), Kyutai SVAD (Kyutai), Mimi Endpointer (Udupa et al., 2025)); and **turn-taking models** (VAP (Ekstedt and Skantze, 2022), MuVAP (Qi and Skantze, 2026), DualTurn (Rajaa, 2026), WavLM-Large anchor (Chen et al., 2022), Turn-1-Mini (p99lab, 2026), ESPnet Turn-Taking (Arora et al., 2025b)). VAP is adapted to multi-party audio through speaker-versus-rest inference. We report EOT and INT performance across speaker-count and interaction-type breakdowns using development-selected operating points. See Appendix E.1 and F.4 for model configurations and score mappings.

Full Duplex Agent Evaluation. We test five full-duplex voice agents—**GPT-Live-1** (OpenAI, 2026), **Gemini-3.8-Live** (Google, 2026), MiniCPM-o-4.5 (Cui et al., 2026), **Realtime-Venus** (Zhao et al., 2026a), and **Step-Audio3-Realtime** (Lin et al., 2026a)—on 120 interactive floor-control scenarios, repeated three times per system. These experiments distinguish offline event detection from agents' observable decisions to enter or relinquish the conversational floor. See Appendix E for scoring configurations.

6 Results

RQ1: Can Models Reliably Identify Floor Transitions?

Turn completion is comparatively tractable; successful floor acquisition is not. Table 3 reveals a pronounced asymmetry between the two floor-control decisions. VAP and Kyutai SVAD achieve comparable EOT recall (92.6% and 92.5%) at low FPR (7.5% and 4.3%), yet recover only 11.7% and 56.2% of successful interruptions, respectively. Even the strongest qualified INT detector misses nearly half of genuine takeovers. Thus, recognizing when a speaker has finished does not translate into determining whether incoming speech has actually displaced the current floor holder.

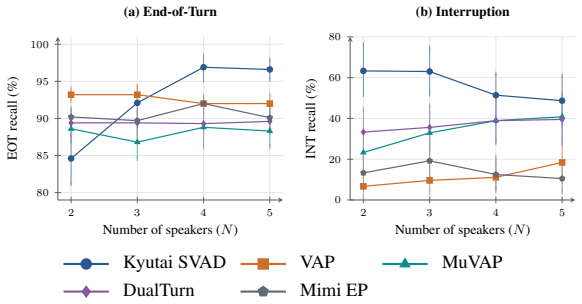


Figure 4: **Speaker-count sensitivity of EOT and INT detection.** Recall by speaker count ($N = 2$–5), with 95% conversation-bootstrap CI. EOT is broadly stable except for Kyutai, whereas INT trends differ. At five speakers, no plotted detector reaches 50% INT recall, highlighting the challenge of identifying successful floor takeovers among multiple speech claimants.

High recall without selectivity is insufficient for conversational control. Acoustic detectors illustrate why floor transitions cannot be reduced to speech activity. Silero VAD achieves 97.2% EOT and 85.4% INT recall, but incurs prohibitive FPRs of 93.5% and 89.2%. Semantic and turn-taking models suppress these false activations, often at the cost of missed takeovers: Mimi Endpointer achieves 90.5% EOT recall but detects only 13.9% of interruptions. This exposes the central challenge: distinguishing genuine floor claims from listener activity that leaves floor ownership unchanged.

Detection quality and latency trade-offs. Even among strong EOT detectors, comparable recall conceals substantial timing differences. VAP detects turn endings at 143 ms p50, versus 1,829 ms for Kyutai SVAD, despite nearly identical recall. Thus, a correct floor transition detected too late may still disrupt natural interaction.

RQ2: Does Speaker Count Make Floor Control Harder?

More speakers do not produce uniform degradation in EOT detection. Figure 4 shows that EOT detection remains comparatively stable across two to five speakers. VAP maintains 92–93% recall, while MuVAP and DualTurn show similarly small variations. Surprisingly, Kyutai improves from 84.6% to 96.6%, suggesting that increasing speaker count does not inherently make turn completion harder under MPTB's controlled conditions.

System	End-of-Turn (EOT)			Interruption (INT)		
	Recall (%) ↑	FPR (%) ↓	p50 (ms) ↓	Recall (%) ↑	FPR (%) ↓	p50 (ms) ↓
Acoustic VAD						
RMS Energy	49.8 [44.4, 57.3]	40.9	292	66.2 [61.5, 72.8]	25.0	859
WebRTC VAD (Google WebRTC Project)	91.8 [90.8, 92.8]	90.3	332	84.3 [80.6, 89.2]	83.8	128
Silero VAD (Silero Team, 2024)	97.2 [96.3, 97.8]	93.5	397	85.4 [81.4, 89.7]	89.2	119
Semantic endpointers						
Smart Turn v2 (Pipecat AI, 2025)	82.5 [81.0, 84.2]	11.8	-18	67.6 [61.5, 73.6]	64.3	87
Kyutai SVAD (Kyutai)	92.5 [91.1, 93.9]	4.3	1829	**56.2** [50.5, 62.0]	6.6	689
Mimi Endpointer (Udupa et al., 2025)	90.5 [89.8, 91.2]	3.8	1231	13.9 [10.0, 17.9]	3.8	870
Turn-taking models						
VAP (Ekstedt and Skantze, 2022)	**92.6** [92.0, 93.1]	7.5	**143**	11.7 [8.1, 15.6]	2.8	492
MuVAP (Qi and Skantze, 2026)	88.1 [87.1, 89.1]	10.2	553	34.5 [29.1, 39.9]	8.6	618
DualTurn (Rajaa, 2026)	89.4 [88.6, 90.2]	10.8	777	37.0 [31.6, 42.6]	8.2	572
WavLM-Large (anchor) (Chen et al., 2022)	90.4 [89.5, 91.2]	22.6	1185	12.8 [9.3, 16.4]	7.3	**42**
Turn-1-Mini (p99lab, 2026)	86.9 [85.9, 87.9]	25.3	1087		—	
ESPnet Turn-Taking (mix) (Arora et al., 2025b)	86.9 [85.6, 88.0]	12.4	1084	1.1 [0.0, 2.5]	1.9	71

Table 3: **EOT and INT detection on `MultiPartyTurnBench`.** Recall is reported with 95% conversation-bootstrap CI; p50 denotes median signed latency (ms). Gray indicates operating points satisfying dev FPR $\leq 10\%$ and test FPR $\leq 15\%$; cyan highlights the highest qualified recall in each task. Budget-qualified systems achieve up to 92.6% EOT recall, yet the best INT recall reaches only 56.2%. Acoustic detectors recover more events but incur substantially higher FPR, highlighting the difficulty of distinguishing genuine floor transitions from speech activity.

Voice agent	START: enter the floor			STOP: yield selectively			
	Start ↑	False start ↓	bACC ↑	Yield ↑	False yield ↓	bACC ↑	$A_{\text{Bias}} \to 0$
GPT-Live-1 (OpenAI, 2026)	93.3	1.7	**95.8** [90.8, 99.2]	50.0	40.6	54.7 [45.7, 63.5]	-0.09
Gemini-3.8-Live (Google, 2026)	88.9	0.0	94.4 [90.0, 97.8]	100.0	97.7	51.1 [50.0, 52.8]	+0.98
MiniCPM-o-4.5 (Cui et al., 2026)	86.7	5.6	90.6 [85.0, 95.6]	1.4	0.0	50.7 [50.0, 52.2]	-0.99
Realtime-Venus (Zhao et al., 2026a)	81.1	8.9	86.1 [80.6, 91.1]	8.1	12.7	47.7 [43.1, 52.4]	-0.79
Step-Audio3-Realtime (Lin et al., 2026a)	0.0	0.0	50.0 [50.0, 50.0]	9.1	15.4	46.9 [34.6, 60.0]	-0.76

Table 4: **Full-duplex agents know when to start, but not when to stop.** START distinguishes genuine turn endings from hold pauses; STOP distinguishes interruptions from backchannels. Four agents achieve 86.1–95.8% START balanced accuracy (bACC), yet all five remain near chance on STOP (46.9–54.7%). Gemini yields to 97.7% of backchannels, whereas MiniCPM-o yields to only 1.4% of interruptions, exposing opposite failure behaviors. Brackets denote 95% scenario-clustered bootstrap CIs; cyan highlights above-chance START performance; red marks STOP estimates whose CIs include 50%.

Interruption detection exhibits opposing speaker-count trends. Kyutai's INT recall declines from 63.3% to 48.7% as speaker count increases from two to five, whereas MuVAP improves from 23.3% to 40.8%, with DualTurn showing a similar upward trend. Yet no model exceeds 50% INT recall at five speakers. These opposing trajectories suggest that models exploit multi-speaker context differently when interpreting competing floor claims. Relational speaker modeling may offer advantages over channel-local endpointing. Speaker count therefore exposes model-specific vulnerabilities rather than a universal scaling penalty. See Appendix F.3 for recall–FPR breakdowns across speaker counts.

RQ3: Which Interaction Types Expose Failures?

Interruption detection is substantially more sensitive to interaction style. Figure 5 (and Appendix F.2) shows that EOT detection remains comparatively stable across interaction types with limited variation. In contrast, INT recall shows large swings under the same threshold (Kyutai SVAD is 47.0% in collaborative dialogue to 77.5% in instructional exchanges). This disparity suggests that EOT detection generalizes more consistently across interaction regimes compared to INT detection.

Conversational difficulty is model-dependent. Kyutai struggles most with collaborative exchanges, while MuVAP performs worst on narrative dialogue (25.0% INT recall), DualTurn on task-oriented interactions (32.6%), and VAP and Mimi Endpointer on argumentative exchanges. These contrasting failures suggest that models rely on different conversational cues when interpreting floor claims, rather than a shared hierarchy of interaction difficulty. See Appendix I for qualitative examples.

RQ4: Can Voice Agents Yield Selectively?

Knowing when to speak does not imply knowing when to stop. Table 4 extends offline detection to observable behavior in five full-duplex agents. Four systems achieve 86.1–95.8% START balanced accuracy, reliably distinguishing genuine

(a) EOT recall (%) ↑

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy†	56.2	47.3	45.6	54.7	44.1	52.6
WebRTC VAD†	90.6	90.5	92.1	92.6	91.7	93.5
Silero VAD†	97.0	96.8	97.1	97.3	97.3	97.8
Smart Turn v2†	81.3	80.4	83.2	83.8	83.4	83.8
Kyutai SVAD	95.1	91.0	90.5	94.0	96.0	90.6
Mimi Endpointer	90.4	92.2	89.2	88.7	90.4	91.2
VAP	92.9	93.7	92.5	91.5	91.2	93.0
MuVAP	86.2	88.9	89.4	87.4	88.0	88.3
DualTurn	89.2	90.0	89.2	87.4	89.6	90.6
WavLM Anchor†	91.1	91.2	88.8	89.6	91.2	90.4
Turn-1-Mini†	89.2	84.9	86.3	86.0	89.0	87.1
ESPnet Turn-Taking	87.4	89.4	86.5	84.1	82.9	88.8

(b) INT recall (%) ↑

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy†	61.8	71.1	65.2	75.0	53.1	69.8
WebRTC VAD†	89.1	91.1	78.8	85.0	75.0	86.0
Silero VAD†	89.1	93.3	83.3	82.5	75.0	86.0
Smart Turn v2†	67.3	73.3	56.1	70.0	84.4	65.1
Kyutai SVAD	54.5	53.3	47.0	77.5	59.4	53.5
Mimi Endpointer	9.1	15.6	15.2	15.0	15.6	14.0
VAP	7.3	15.6	9.1	15.0	9.4	16.3
MuVAP	30.9	37.8	34.8	35.0	25.0	41.9
DualTurn	36.4	40.0	33.3	47.5	34.4	32.6
WavLM Anchor	12.7	6.7	21.2	7.5	15.6	9.3
Turn-1-Mini	—	—	—	—	—	—
ESPnet Turn-Taking	0.0	0.0	1.5	2.5	0.0	2.3

Figure 5: **EOT / INT recall by interaction type on `MultiPartyTurnBench` test set.** Darker means higher recall. † denotes over the FPR budget; its recall reflects indiscriminate firing. EOT recall remains comparatively stable across conversation types. INT recall swings up to 30 points. Multi-party floor-control difficulty depends on conversational dynamics (argumentative for VAP and Mimi EP, collaborative for Kyutai, narrative for MuVAP, and task-oriented for DualTurn), not merely on speaker count.

turn endings from mid-turn holds. Yet this competence collapses when deciding whether to yield: all five achieve only 46.9–54.7% STOP balanced accuracy, with confidence intervals including chance performance. The offline EOT–INT asymmetry thus manifests directly in agent behavior: models recognize end of turn but struggle to determine when incoming speech interruptions.

Similar STOP accuracy conceals opposite behavioral failures. Gemini-3.8-Live effectively surrendering the floor indiscriminately (97.7% of backchannels, bias $+0.98$). Conversely, MiniCPM-o-4.5 almost never yields, ignoring 98.6% of genuine interruptions (bias -0.99). Realtime-Venus similarly resists yielding, while GPT-Live-1 is more balanced but remains unreliable. These opposing behaviors suggest inability of the voice agents to distinguish the conversational significance of incoming audio (see Appendix G).

7 Discussion

Multi-party floor control requires reasoning about speaker roles, not merely speech activity. MPTB exposes a fundamental limitation of conventional endpointing: incoming speech does not necessarily imply a change in floor ownership. Reliable interruption handling in multi-speaker settings requires distinguishing competing floor claims from backchannels and other non-floor-taking activity, conditioned on the current speaker and conversational context. This motivates event-conditioned modeling of floor transitions rather than independent speech-activity decisions.

Floor-control evaluation must capture the consequences of incorrect decisions. High event recall can conceal excessive false activations, while conservative yielding policies can appear reliable despite missing genuine interruptions. MPTB demonstrates the importance of evaluation breakdowns across speaker counts and interaction regimes. Such diagnostics expose failure modes obscured by aggregate scores and provide actionable targets for developing more selective full-duplex systems.

Synthetic to real evaluation gap. Table 24 illustrates that this difficulty is not confined to synthetic speech: VAP recovers only 35.9% of interruptions on real four-party AMI meetings at 5.3% FPR, despite in-slice threshold tuning. This supports corroborating learnings from synthetic evaluations to real-world relevance of the interruption challenge.

8 Conclusion

We introduced `MultiPartyTurnBench`, a controlled benchmark for turn taking dynamics for end-of-turn and interruption detection in multi-party spoken dialogues. Existing systems detect turn completion reliably but struggle to recognize floor takeovers without excessive false activations, with failures varying across conversational regimes and speaker counts. This asymmetry extends to full-duplex agents where all systems perform near chance on selective yielding. `MultiPartyTurnBench` establishes selective interruption handling as a critical unresolved challenge for reliable multi-party voice interaction.

9 Limitations

Synthetic speech and external validity. MPTB's controlled construction provides exact floor-state annotations and reproducible interaction conditions, but synthesized speech cannot fully capture the prosody, disfluencies, timing adaptations, and ambiguous floor negotiations of spontaneous conversation. The synthetic-to-real comparison (Table 24) reveals non-uniform difficulty: learned endpointers generally achieve higher EOT recall on MPTB, whereas VAP detects fewer interruptions on MPTB (11.7%) than on real four-party AMI (35.9%). Thus, synthetic evaluation does not uniformly overestimate performance, and MPTB scores should be interpreted as controlled diagnostics rather than estimates of natural-conversation error rates.

Scope of real-speech validation. Our multi-party validation uses 15 four-speaker AMI meetings with floor events derived from manual speech segments rather than independently annotated under the MPTB taxonomy. The evaluated corpora also differ in speaker count, conversational domain, acoustics, annotation protocols, and threshold calibration, with real-audio slices using in-slice tuning. Cross-corpus differences therefore cannot be attributed causally to speaker count or synthetic speech. AMI provides complementary evidence that interruption detection remains challenging in natural multi-party conversation, but does not establish direct synthetic-to-real performance transfer.

Linguistic and demographic coverage. MPTB is restricted to English, with accent diversity limited to the voices sampled during synthesis rather than representative real-world speaker populations. We do not systematically evaluate generalization across languages, regional accents, or speaker demographics. Turn-taking conventions, prosodic cues, and interruption behaviors may vary across linguistic and sociocultural contexts. Consequently, our findings should not be interpreted as universal patterns of conversational floor control; broader multilingual and demographically diverse real-speech evaluation is necessary to establish their generality.

Ethics Statement

The released MPTB corpus contains synthetic speech and fictional LLM-generated dialogue rather than newly collected private conversations. The benchmark is designed to diagnose conversational control failures and should not be interpreted as a safety certification for deployment. Real Turn-Bench material used for scorer validation is not redistributed, and AMI is used under its existing research terms. Because full-duplex systems may be used in multi-user or high-stakes settings, benchmark performance should be considered alongside broader evaluations of recognition quality, safety, fairness, privacy, and task correctness.

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A Human Evaluation Study

A.1 Human Annotation

Although MPTB derives floor-state labels before speech synthesis, a procedurally valid event may not be perceived as intended in the rendered audio. We therefore conduct a human validation study assessing whether listeners recognize the intended event categories and their temporal realization. The study covers all 240 conversations and 6,116 candidate events, with three independent annotations per event (18,348 primary categorical judgments, excluding blind repeats).

Annotators and qualification. Annotators have native or near-native English comprehension and complete written guidelines, worked audio examples, and a held-out calibration set emphasizing turn completion versus continuation, backchannels versus floor claims, and successful versus unsuccessful interruptions. Qualification requires at least 75% agreement with expert labels across 10 calibration examples. Annotators are blinded to generator labels, event logs, and model predictions.

Annotation procedure. Annotators inspect candidate regions with 10 s of preceding and following context using synchronized multi-speaker audio, independently controllable speaker channels, and a shared timeline. For each region, they assign an event category, identify the perceived floor holder before and after the event, and mark onset and offset timestamps. An AMBIGUOUS label is permitted when the floor outcome cannot be determined reliably. Automatic transcripts are available as references, but audio takes precedence. Table 5 summarizes the annotation fields.

Annotations are collected independently without discussion or adjudication. Examples are presented in randomized order across interaction conditions, speaker counts, and voices. A subset of 25% of examples is repeated blindly to assess intra-annotator consistency. Since listeners receive subsequent conversational context, this study validates the perceptual realization of floor transitions rather than their causal predictability at event onset.

Field	Values recorded per candidate region
Event category	TURN-YIELD, TURN-HOLD, INTERRUPTION, NON-FLOOR-TAKING INT., BACKCHANNEL, NONCONTENT, AMBIGUOUS
Floor holder (pre)	Perceived speaker ID immediately before the event $(1, \ldots, N)$
Floor holder (post)	Perceived speaker ID immediately after the event $(1, \ldots, N)$
Onset / offset	Timestamps (s) on the shared timeline
Stimulus	Noisy speaker channels (24 kHz), solo/mix controls, speaker timeline, reference transcript
Hidden information	Generator label, event log, model predictions, condition metadata

Table 5: **Human annotation schema.** Annotators independently identify event categories, floor holders, and temporal boundaries using the role-relative taxonomy (Table 13). NONCONTENT corresponds to listener laughter in the released corpus.

A.2 Consensus and Agreement Measures

We construct human consensus independently of the generator event log. A candidate receives a majority-consensus label when at least two of three annotators assign the same event category. Candidates without categorical consensus are treated as unresolved for human-validation analysis. For agreed events, the consensus boundary is the median timestamp of the agreeing annotators.

We distinguish three complementary measures. **Categorical consensus** is the fraction of candidate events receiving a majority label, while Fleiss' κ measures inter-annotator categorical agreement corrected for chance. **Boundary F1** measures temporal agreement using a 500 ms boundary tolerance. Finally, **generator–human agreement** measures whether the procedurally assigned event label matches the independent human consensus. Fleiss' κ is calculated using a one-versus-rest measure over the complete corpus. Boundary F1 temporal agreement is calculated relative to consensus boundaries.

Let $\mathcal{C}$ denote events with categorical consensus, and let y_i^{gen} and y_i^{human} be the generator and consensus labels. Generator–human agreement is

$$A_{\text{gen}} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \mathbf{1}\left[y_i^{\text{gen}} = y_i^{\text{human}}\right]. \qquad (1)$$

Thus, generator–human agreement is conditional on human consensus and should not be interpreted as accuracy over all generated events.

A.3 Human Validation Results

Floor-event labels are perceptually distinguishable, but not equally unambiguous. Table 6 reports agreement across the six generated event categories. Backchannels show the strongest consistency, with 93.6% categorical consensus, Fleiss' $\kappa = 0.92$, and boundary F1 of 0.95. Successful interruptions exhibit lower consensus (81.4%) and temporal agreement (F1 $= 0.89$), indicating greater uncertainty in identifying the realization and timing of floor takeovers. Turn endings and mid-turn

holds also exhibit lower categorical consensus than backchannels, supporting the distinction between acoustic speech cessation and perceived floor relinquishment.

Generator–human agreement supports label plausibility, with important qualifications. Among events receiving human consensus, generator–human agreement ranges from 86.5% for turn endings to 97.8% for backchannels. Successful interruptions achieve 91.2% generator–human agreement despite having the lowest categorical consensus (81.4%). These measures capture different properties: consensus reflects consistency across annotators, whereas generator–human agreement evaluates label correspondence only on retained events. High conditional agreement therefore supports the plausibility of recognized synthetic events but does not establish that every generated event is perceptually unambiguous.

Human agreement decreases with speaker count. Table 7 shows a consistent decline across all three agreement measures as the number of speakers increases. Categorical consensus falls from 87.8% at $N = 2$ to 80.1% at $N = 5$, while Fleiss' κ decreases from 0.84 to 0.76 and boundary F1 from 0.96 to 0.90. The simultaneous reduction in categorical and temporal agreement suggests that resolving floor transitions becomes less consistent across annotators in larger groups. This is a descriptive association rather than an isolated causal effect of speaker count, since realized event composition and acoustic conditions may also differ.

Interpretation and scope. The human study provides complementary evidence that MPTB's generated floor-state categories have recognizable acoustic and conversational realizations, while identifying residual ambiguity in floor transitions, particularly successful interruptions. Since candidate regions are selected from the generator event log and annotators may inspect future context, these findings validate event-label plausibility rather than independent event discovery or real-time detectability. The study also does not establish that synthetic speech captures the full variability of natural multiparty interaction.

B MultiPartyTurnBench Evaluation Tracks

Table 11 summarizes the two benchmark tracks and their positive and hard-negative events. This section gives the exact anchor derivation and scoring rules behind every number in the paper.

B.1 Anchor Derivation

Scoring anchors are derived deterministically from the canonical event log, separately for every speaker channel s:

- **EOT positives**: TURN events of s with outcome *yield*; anchor t = segment offset. The last turn of a conversation is a yield.
- **EOT hard negatives**: TURN events of s with outcome *hold*; scored span = [pause onset, same-speaker resume onset].
- **EOT excluded**: TURN events of s with outcome *interrupted*; a ± 0.25 s neighborhood around the forced stop is neither rewarded nor penalized.
- **INT positives**: CLAIM_FLOOR events of s resolved as INTERRUPTION; anchor t = claim onset on the interrupter's own channel.
- **INT hard negatives**: VOCALIZE_NO_CLAIM events of s (backchannel or laughter); scored span = [onset, offset].
- **INT excluded**: CLAIM_FLOOR events of s resolved as NON-FLOOR-TAKING INTERRUPTION; predictions inside [onset, offset] are discarded before matching.

Acoustic nuisance events (ambient bed, MUSAN sound effects, distant third-party speech) never generate anchors. Predictions they trigger outside scored spans are ignored, so they affect a system only when they fall inside a negative span or preempt a positive.

B.2 Matching, Metrics and Operating Points

Each system submits, per speaker channel, causal decision times for each track. Decision times are rising edges of a score crossing threshold θ, followed by a 2 s refractory period. Matching follows TurnBench (Jiang et al., 2026): (1) predictions inside excluded spans are dropped; (2) gold positives are visited in time order and each is matched to the earliest unclaimed prediction in $[t-0.25,\ t+3.0]$ s (TP, else FN); (3) each hard-negative span counts as one FP if any remaining unclaimed prediction falls inside it, otherwise one TN. Signed latency is $t_{\text{pred}} - t_{\text{gold}}$ for matched pairs. Metrics are pooled over all speakers and conversations of a split.

For every system and track, θ is chosen on the development split as the value maximizing recall

Event Class	# Events	Consensus (%)	Fleiss' κ	Boundary F1	Gen.–Human Match (%)
TURN / Yield	3,422	83.4	0.80	0.94	86.5
TURN / Hold	241	85.6	0.84	0.93	87.3
INTERRUPTION	367	81.4	0.78	0.89	91.2
NON-FLOOR-TAKING INT.	605	88.6	0.82	0.91	92.5
BACKCHANNEL	1,176	93.6	0.92	0.95	97.8
NONCONTENT	305	90.5	0.83	0.90	94.5

Table 6: **Human validation of MPTB floor-state events.** Event counts cover all 240 conversations, with three independent annotator judgments per candidate. Consensus denotes majority categorical agreement; Fleiss' κ measures chance-corrected inter-annotator agreement; boundary F1 measures temporal agreement within 500 ms; and generator–human match is conditional on consensus (Eq. 1). NONCONTENT denotes listener laughter. Non-floor-taking interruption attempts are included in human validation but excluded from causal INT scoring. Pooled agreement statistics require event-level recomputation and are not estimated by averaging class-level coefficients.

	Corpus		Human Validation		
N	Conv.	Events	Cons. (%)	Fleiss' κ	Bnd. F1
2	60	1,490	89.8	0.84	0.96
3	60	1,527	88.6	0.82	0.93
4	60	1,536	84.3	0.81	0.91
5	60	1,563	82.1	0.76	0.90

Table 7: **Human agreement across speaker counts.** Event totals include the six generator-defined categories in Table 6. Categorical consensus, Fleiss' κ, and boundary F1 consistently decrease from two to five speakers.

subject to $\mathrm{FPR}_{\mathrm{dev}} \leq 0.10$. When no candidate qualifies, the candidate with the lowest development FPR is selected (ties broken by recall). The selected θ is frozen for the whole test split and for all subgroup analyses. A system is *in budget* when $\mathrm{FPR}_{\mathrm{dev}} \leq 0.10$ and $\mathrm{FPR}_{\mathrm{test}} \leq 0.15$. Confidence intervals are percentile intervals from a conversation-clustered bootstrap ($B = 10{,}000$, seed 20261010); paired system differences resample the same conversations for both systems. Table 12 lists the scored denominators.

C Benchmark Construction Details

C.1 Role-relative event taxonomy

Every event is defined relative to the current floor holder. Table 13 gives the operational meaning of each class, its scoring role, and its realized frequency in the released corpus. Labels are produced by the generator (Stage 1 proposes the symbolic act; Stage 2 resolves its outcome) and are never inferred from audio.

C.2 Interaction-Type Definitions

MPTB adopts the six conversation types used in TurnBench (Jiang et al., 2026), which were designed to induce distinct patterns of turn-taking behavior. The categories are retained to provide a common interaction-style axis with prior work, while MPTB uses them within a balanced multi-party factorial design over $N \in \{2, 3, 4, 5\}$. Table 14 summarizes the intended interaction characteristics used to condition synthetic dialogue generation. These descriptions specify the dominant conversational regime rather than enforcing deterministic event patterns; realized backchannel, interruption, overlap, and turn-duration statistics are reported separately in Appendix D.1. Table 15 gives the exact style string and soft targets passed to the generator for each type, and Figure 6 shows one excerpt per type from the test split.

C.3 Generation pipeline

Generation proceeds in five stages. (1) An LLM writes a role-tagged dialogue conditioned on party count, interaction type, and target rates for backchannels and interruptions. (2) A deterministic state machine resolves floor claims, holds, yields, and overlaps into a timed event graph. (3) Kokoro-82M renders each speaker using a disjoint voice pool. (4) A controlled noise stage adds procedural ambient sound, discrete non-content events, and intermittent distant speech. (5) The assembler emits synchronized per-speaker audio and the exact event log. Scripts, seeds, audio manifests, and scorer inputs are recorded for reproducibility. The remainder of this section specifies every stage.

Conversation specifications. Specifications are sampled with exactly 10 conversations per (N, type) cell (2 dev, 8 test). Each specification independently draws a domain (20 domains, Table 16), a tone (5 tones), a target duration $D \sim \mathcal{U}(150, 240)$ s, N distinct voices from the split's voice pool, and a noise profile (Stage 4). Each conversation carries an integer seed that drives every

Interaction Type	# Conv.	Hours	Turns/min	BC/min	INT/min	Overlap (%)	Mean Turn (s)
Casual	40	1.10	12.7	4.24	0.92	11.4	4.81
Task-Oriented	40	1.03	11.4	3.80	0.89	9.9	5.27
Instructional	40	1.17	7.9	3.81	0.70	10.7	7.79
Collaborative	40	1.07	11.4	3.43	1.33	11.0	5.32
Argumentative	40	1.11	9.6	2.95	1.08	10.7	6.29
Narrative	40	1.41	6.6	3.34	0.53	9.1	9.22
All	**240**	**6.9**	**9.8**	**3.58**	**0.89**	**10.4**	**6.45**

Table 8: **Corpus statistics by interaction type.** Turns/min counts floor-holding TURN segments; BC/min includes backchannels and laughter; INT/min counts successful interruptions; overlap denotes the fraction of conversation time with simultaneous vocal activity from at least two speakers. Interaction types induce distinct conversational regimes despite balanced conversation counts.

N	# Conv.	EOT	INT	BC/min	Overlap (%)
2	60	853	80	3.12	8.9
3	60	838	96	3.61	10.9
4	60	875	94	3.60	10.9
5	60	856	97	4.06	11.1
All	**240**	**3,422**	**367**	**3.58**	**10.4**

Table 9: **Corpus composition by speaker count.** Each party-count condition contains 60 conversations. EOT and INT denote positive event anchors; overlap measures the percentage of conversation time containing simultaneous vocal activity.

Split	# Conv.	Hours	# Voices	EOT / INT
Dev	48	1.42	8	706 / 86
Test	192	5.47	16	2,716 / 281
Total	**240**	**6.89**	**24**	**3,422 / 367**

Table 10: **MPTB development and test partitions.** The partitions use disjoint TTS voices and retain the complete speaker-count × interaction-type design, with two development and eight test conversations per condition. EOT and INT denote positive event counts.

downstream random draw.

Stage 1: event-conditioned dialogue generation. Scripts are generated by `gpt-4.1` (all 240 conversations) with the system prompt and user template of Figures 7 and 8, temperature 0.9, a 8,000-token output limit and JSON output. Each utterance carries exactly one event type (TURN, CLAIM_FLOOR, VOCALIZE_NO_CLAIM), a qualitative timing hint and, where applicable, a target holder and a resolution hint. Outputs are validated against structural rules (first utterance is a TURN; claims and vocalizations require a current holder and target it; every speaker holds at least one TURN; at least $\max(2N, 8)$ turns; closed tag and hint vocabularies). A failed validation is returned to the model with the error list for repair, up to three attempts with seeds seed $+ k$. Before validation, any backchannel longer than three words is deterministically relabelled as a non-floor-taking claim, so that multi-word listener speech is never scored as an INT negative. Realized scripts contain 6,800 utterances (28.3 per conversation): 4,347 TURN (median 16 words), 367 successful claims (median 9 words), 605 failed claims (median 5 words), 1,176 backchannels (median 1 word) and 305 laughter vocalizations (median 2 words).

Stage 2: floor-state orchestration. A discrete-event state machine consumes the script in order and maintains the floor holder. A TURN by a new speaker closes the previous holder's segment as a *yield* and starts the new segment after a gap drawn from the timing hint. A TURN by the current holder continues the segment; if the open segment is at least 4 s long, a mid-turn pause is inserted with probability 0.45, closing the segment as a *hold* and producing an EOT hard negative. A successful CLAIM_FLOOR starts inside the holder's ongoing utterance, the holder keeps talking briefly in overlap and is then truncated (*interrupted*), and the floor transfers. A failed claim and a vocalization overlap the holder without changing the floor. Utterance durations come from the synthesized audio, so timing is resolved against real speech lengths. Table 17 lists every distribution; all draws use the conversation seed.

Stage 3: speaker rendering. Each utterance is synthesized independently with Kokoro-82M (hexgrad, 2024) (`hexgrad/Kokoro-82M`) at 24 kHz, speed 1.0, with a per-utterance seed derived from the text and voice. Leading and trailing silence below -40 dBFS is trimmed with a 30 ms margin, and clips are cached by content hash so re-rendering is byte-reproducible. Abstract voice ids

Track	Positive: floor changes	Hard negative: floor unchanged	Prediction anchor
EOT	Current floor holder completes a turn and yields the floor.	Mid-turn pause after which the same speaker resumes while retaining the floor.	End of floor-holding speech
INT	Listener begins speaking while another speaker holds the floor and successfully takes ownership.	Backchannel or non-content listener activity that does not change floor ownership.	Onset of floor-taking speech

Table 11: **`MultiPartyTurnBench` evaluation tracks.** Each track contrasts a floor-changing event with a conversationally similar event that leaves floor ownership unchanged. This paired formulation prevents systems from succeeding through indiscriminate silence or speech-activity detection.

		EOT		INT	
Split	**Conv.**	**Pos.**	**Neg.**	**Pos.**	**Neg.**
Dev	48	706	55	86	309
Test	192	2,716	186	281	1,172
Total	240	3,422	241	367	1,481

Table 12: **Scored denominators** (positive anchors and hard-negative spans). Every test-split number in the paper is computed over these anchors. INT negatives comprise 1,176 backchannels and 305 laughter vocalizations (test: 919 and 253); the 605 non-floor-taking claims are excluded intervals.

Event	Operational meaning	Role	Count
TURN–yield	Holder releases the floor to another speaker (or the conversation ends).	EOT +	3,422
TURN–hold	Holder pauses mid-turn and resumes.	EOT −	241
TURN–interrupted	Holder is cut off by a successful interruption.	EOT excl.	367
INTERRUPTION	Listener takes the floor before the current turn naturally ends.	INT +	367
NON-FLOOR-TAKING INT.	Listener attempts a claim but the holder keeps the floor.	INT excl.	605
BACKCHANNEL	Listener vocalizes ($\leq$3 words) without claiming the floor.	INT −	1,176
NONCONTENT	Listener laughter without claiming the floor.	INT −	305

Table 13: **Role-relative event taxonomy** used by the generator and scorer, with counts over all 240 conversations. Acoustic nuisance (ambient bed, sound effects, distant speech) is rendered in Stage 4 and carries no label.

map to Kokoro voices as in Table 18. The 24 voices are split gender-balanced into 8 development and 16 test voices; no voice occurs in both splits, and no voice is reused within a conversation.

Stage 4: acoustic scene rendering. Clips are placed on per-speaker channels at the orchestrated times; truncated holder clips receive a 30 ms fade, and each channel is peak-normalized to -1 dBFS. A single noise realization is added to all channels of a conversation (one room, N microphones), so nuisance sounds are cross-channel. Table 19 gives the noise model and its realized distribution; sound effects and distant speech are drawn from MUSAN (Snyder et al., 2015). If the mixture exceeds -1 dBFS, one common gain is applied to all channels, preserving inter-channel levels.

Stage 5: channel assembly, quality control and release. Each conversation is released as N noisy evaluation channels and N clean channels (FLAC, 24 kHz), the tagged script, the canonical event log, a render sidecar listing every clip with its placement and truncation, the noise manifest, and the seed. An automatic acoustic check runs on the clean channels: (Q1) audio length matches the event log; (Q2) per-channel speech ratio in $[0.03, 0.99]$; (Q3) at least 50% of each scheduled clip span is voiced; (Q4) at most 30% voiced time outside scheduled clips; (Q5) at least 60% of overlap events (interruptions, failed claims, vocalizations) have both mover and holder voiced. All 240 conversations pass. Across the corpus, 2,369 of 2,436 overlap events (97.2%) are acoustically realized as simultaneous speech, median clip coverage is 0.81, and no channel shows speech leakage outside its clips.

From script to scoring anchors · test_00013 (N=5, narrative)

Event log (first 78 s) → anchors on each speaker channel.

Time (s)	Spk	Generator event	Anchor
0.30–19.19	S1	TURN, yield→S2	EOT+ @19.19 (S1)
7.97–9.18	S2	BACKCHANNEL "Oh, absolutely."	INT− [7.97, 9.18] (S2)
9.45–9.90	S3	BACKCHANNEL "mm-hm"	INT− [9.45, 9.90] (S3)
14.88–16.21	S4	failed claim "Yeah, that's a good one."	INT excl. (S4)
19.21–44.33	S2	TURN, hold	EOT− [44.33, 45.29] (S2)
34.70–35.38	S3	failed claim "Right, I—"	INT excl. (S3)
45.29–51.98	S2	TURN, yield→S3	EOT+ @51.98 (S2)
51.94–77.85	S3	TURN, interrupted	EOT excl. @77.85±0.25
77.03	S5	INTERRUPTION "I, uh—can I just say …"	INT+ @77.03 (S5)

Reading. S2 takes the floor 22 ms after S1 stops (a `latched` hand-over). The S2 hold is created by a same-speaker continuation after S3's failed claim; its scored span is the 0.96 s silence before S2 resumes. S5's claim starts 0.82 s before S3 is cut off, so the interruption onset, not S3's forced stop, is the scored event.

Interaction Type	Conversation Setting	Expected Floor-Management Characteristics
Casual	Unstructured social conversation with flexible topic progression and informal exchange.	Frequent acknowledgements and backchannels, relatively short turns, topic shifts, and occasional overlap.
Task-Oriented	Goal-directed interaction in which participants jointly complete or resolve a concrete task.	Frequent clarification and confirmation, comparatively structured floor transfer, and short coordination turns.
Instructional	Asymmetric interaction in which one participant primarily explains, teaches, or guides others.	Longer floor holds by the primary speaker, question-driven transitions, and acknowledgement backchannels from listeners.
Collaborative	Shared reasoning or problem solving without a single dominant speaker.	More frequent floor exchange, cooperative overlap, short claims to the floor, and relatively balanced participation.
Argumentative	Structured disagreement or debate in which participants defend competing positions.	Longer contested turns, more competitive floor claims, interruptions, and overlapping speech.
Narrative	One participant primarily recounts an event or story while others act as listeners.	Longer sustained turns, frequent listener backchannels, and comparatively limited floor competition.

Table 14: **Interaction types used in MultiPartyTurnBench.** MPTB adopts the six conversation categories used in TurnBench (Jiang et al., 2026) and instantiates each across $N \in \{2, 3, 4, 5\}$ speakers. The final column summarizes the intended floor-management regime used to condition dialogue generation; realized event statistics are measured from the generated corpus rather than assumed from these descriptions.

		Soft targets (TurnBench dyadic statistics)			
Type	**Style string inserted into the prompt**	**Turn (s)**	**BC/min**	**INT/min**	**Words/min**
Argumentative	*a heated debate with frequent disagreement, interruptions, and people talking over each other*	10.0	3.09	2.48	206
Casual	*a relaxed casual chat with frequent overlaps, backchannels, and light interruptions*	8.1	5.20	2.05	211
Collaborative	*participants actively building on each other's ideas, with overlaps and frequent floor contests*	7.9	3.97	2.64	207
Instructional	*one participant explaining/teaching others who ask clarifying questions; mostly clean turns, few interruptions*	10.5	4.30	1.52	202
Narrative	*storytelling with personal anecdotes; longer turns, listeners backchanneling, rare interruptions*	10.2	4.76	1.50	198
Task-Oriented	*a focused task-oriented working session; mostly orderly turns with occasional quick interjections*	8.9	5.00	1.75	205

Table 15: **Generator conditioning per interaction type.** For a conversation of target duration D (min) with N speakers, the prompt requests about $\max(2N, \mathrm{round}(60D/\mathrm{turn}))$ floor-holding turns, $\mathrm{round}(\mathrm{INT/min} \cdot D \cdot s_N)$ floor claims and $\mathrm{round}(\mathrm{BC/min} \cdot D \cdot s_N)$ backchannels/laughter, with $s_N = 1 + 0.1(N - 2)$. Targets are guardrails; realized rates are reported in Tables 20 and 8.

Domains (20). tech support call; study group / exam prep; family logistics planning; product design standup; podcast panel / current-events discussion; board game night; job interview; travel itinerary planning; classroom discussion; relationship/interpersonal conflict; restaurant/group dining decision; sports commentary; book club; home renovation planning; medical appointment scheduling (logistics only, non-clinical); roommate household planning; startup pitch practice; book/movie recommendation swap; volunteer event coordination; group trivia night.

Tones (5; realized count). professional and business-like (57); curious and inquisitive (54); enthusiastic and energetic (53); casual and friendly (42); thoughtful and contemplative (34).

Table 16: **Scenario conditioning vocabulary.** All 20 domains occur in the corpus (most frequent: volunteer event coordination, 19 conversations).

D Corpus Composition and Turn-Taking Dynamics

D.1 Benchmark Analysis

Dataset composition. MPTB crosses four speaker counts ($N \in \{2, 3, 4, 5\}$) with six interaction types, yielding 24 conditions with 10 conversations each. The resulting 240 conversations comprise 6.89 hours of speech, partitioned into 48 development and 192 test conversations with disjoint speaker voices. Table 20 provides event counts and conversational statistics for each condition. Although conversation counts are balanced, realized positive and negative event counts vary across conditions.

Argumentative · N=4 · test_00131

S1 [turn] —and some of them can't drive in that kind of storm. If we postpone, it's safer for everyone.
S3 [backchannel] right
S2 [turn] You guys, the forecast changes all the time, though. What if it clears up? We'll lose all the momentum if we change it for no reason—
▷ S4 [interruption] *Can I just—*
S4 [turn] What about a backup plan? Could we have an indoor option or a rain date that we communicate clearly to everyone?

Casual · N=4 · test_00094

S3 [turn] Great. Uh, snack duty—
▷ S1 [interruption] *Actually, let me do snacks! I'll bake something this time.*
S1 [turn] I felt bad bringing just fruit last time.
S2 [non-floor-taking] Fruit was totally fine.

Collaborative · N=4 · test_00167

S3 [turn] Alright, so Thursday at 9:30? Should we just pencil that in or—
▷ S1 [interruption] *Wait, does the lab need to be notified? Because last time—*
S3 [turn] Oh, right, last time the bloods weren't ready.
S4 [turn] Massive plot twist: I actually already cc'd the lab this morning. Gold star for me.

Instructional · N=5 · test_00138

S1 [turn] —if none of the pairs work for you, or you want a better combination, you can spend a nature token to take any tile and animal, not just the paired ones.
S4 [non-floor-taking] Right, right. That makes sense.
S3 [turn] So scoring—uh, is it based on the animals or the terrains, or both?
▷ S5 [interruption] *Like, do we get more points for bigger groups or for having every animal?*
S2 [laughter] haha

Narrative · N=5 · test_00013

S1 [turn] So, uh, I'll just start—last weekend I finally watched *The Last Black Man in San Francisco*. [...] It actually reminded me a lot of when I read Colson Whitehead's *Nickel Boys*, how it just lingers with you for days.
▷ S2 [backchannel] *Oh, absolutely.*
S3 [backchannel] mm-hm
S4 [non-floor-taking] Yeah, that's a good one.

Task-Oriented · N=2 · test_00035

S1 [turn] Okay, so, Tuesday: land, hotel, park, ramen, robot cafe, konbini dinner, sleep. Wednesday: onsen, Hakone by train. That feel like the right flow?
S2 [non-floor-taking] Yeah, yeah, I like it a lot.
▷ S2 [interruption] *Oh, wait—*
S2 [turn] —should we pack swimsuits for the onsen? Or do they give you stuff there?

Figure 6: **One generated excerpt per interaction type** (test split, verbatim script text; tags are the generator's event labels, highlighted lines are scored anchors). Argumentative and collaborative claims cut into unfinished clauses; casual and task-oriented claims are short coordination moves; instructional claims are listener questions; narrative regions are dominated by stacked listener backchannels that must not be scored as takeovers.

Parameter	Value
Lead-in silence	0.3 s
Yield gap, `after_pause`	$\mathcal{U}(0.20, 0.60)$ s
Yield gap, `latched`	$\mathcal{U}(-0.12, 0.06)$ s
Yield gap, `long_pause`	$\mathcal{U}(0.80, 2.00)$ s
Same-holder continuation gap	$\mathcal{U}(0.02, 0.18)$ s
Mid-turn pause: probability / min. segment	0.45 / 4.0 s
Mid-turn pause duration	$\mathcal{U}(0.40, 1.20)$ s
Interruption onset (fraction of holder utterance)	$\mathcal{U}(0.45, 0.85)$
Holder overlap before cut-off	$\mathcal{U}(0.40, 1.20)$ s
Failed-claim onset (fraction)	$\mathcal{U}(0.30, 0.80)$
Backchannel/laughter onset (fraction)	$\mathcal{U}(0.20, 0.80)$
Self-overlap guard (same channel)	0.08 s
Truncation fade-out	30 ms

Table 17: **Stage-2 orchestrator parameters.** Realized timing distributions are given in Table 21.

ID	Kokoro	G	Style	ID	Kokoro	G	Style
v01	af_alloy	F	clipped/efficient	v13	am_onyx	M	upbeat/expressive
v02	af_aoede	F	energetic/upbeat	v14	af_nicole	F	dry-humor/deadpan
v03	am_adam	M	calm/measured	v15	am_puck	M	laid-back/casual
v04	am_echo	M	forceful/assertive	v16	am_santa	M	deadpan/flat
v05	am_eric	M	warm/friendly	v17	af_nova	F	skeptical/probing
v06	am_fenrir	M	hesitant/soft-spoken	v18	bm_daniel	M	nervous/tentative
v07	am_liam	M	thoughtful/slow	v19	af_river	F	gentle/reassuring
v08	af_bella	F	measured/rebuttal	v20	af_sarah	F	authoritative/instructor
v09	af_heart	F	playful/animated	v21	bm_fable	M	gravelly/mature
v10	af_jessica	F	direct/no-nonsense	v22	af_sky	F	bright/youthful
v11	af_kore	F	brisk/professional	v23	bm_george	M	analytical/precise
v12	am_michael	M	moderator/neutral	v24	bf_alice	F	empathetic/attentive

Table 18: **Voice pool and split.** Shaded ids form the development pool (4F/4M); the remaining 16 form the test pool. Prefix a/b = American/British English. The style descriptor is passed to the script generator.

Turn-taking dynamics. Speaker count has a comparatively limited effect on aggregate conversational dynamics: overlap remains between 10.2 and 11.0 seconds per dialogue, while interruption rates increase from 0.70/min at two speakers to approximately 0.97/min at five. Interaction type induces larger differences in conversational behavior. Casual conversations have the fastest exchanges (19.4 events/min) and shortest turns (4.81 s), whereas Narrative conversations exhibit longer turns (9.22 s) and fewer interruptions (0.53/min). Collaborative dialogue has the highest interruption rate (1.33/min). These differences establish distinct floor-management regimes within

Stage-1 system prompt, verbatim (1/2): output schema

```
You are a meticulous dialogue data generator that writes REALISTIC
multi-party spoken conversation transcripts for a turn-taking
benchmark. You output STRICT JSON only.

The transcript is an ordered list of utterances. Each utterance is one
JSON object with these fields:
- "speaker": one of the given speaker ids (e.g. "speaker_1").
- "text": what is actually said aloud. Natural spoken English
(disfluencies, "um", "like", trailing "—" for cut-offs are welcome).
No stage directions, no narration, no lists/bullets.
- "event_type": exactly one of:
  - "TURN": a normal floor-holding utterance (the speaker "has the
floor"). This is the default.
  - "CLAIM_FLOOR": this speaker jumps in to try to take the floor FROM
the current floor-holder while that holder is still speaking. REQUIRES:
    - "resolved_hint": "INTERRUPTION" (the jump-in SUCCEEDS — the
holder stops and this speaker takes over; usually followed by a TURN
from this same speaker) OR "NON_FLOOR_TAKING_INTERRUPTION" (the
jump-in FAILS — it is a brief overlap and the ORIGINAL holder keeps
going; usually the holder's next TURN continues their sentence).
    - "target": the speaker id being interrupted (the current holder).
  - "VOCALIZE_NO_CLAIM": a backchannel or laughter over the current
holder; it NEVER takes the floor. REQUIRES:
    - "sublabel": "BACKCHANNEL" or "LAUGHTER".
    - "target": the current holder id being backchanneled.
    - A "BACKCHANNEL" MUST be an extremely brief vocalization of at
most THREE words — ideally ONE — e.g. "mm-hm", "yeah", "right",
"uh-huh", "exactly", "oh really", "no way". It is a listening noise,
not a comment. If a listener would say a whole phrase or sentence,
that is NOT a backchannel: make it a "CLAIM_FLOOR" instead
(resolved_hint "NON_FLOOR_TAKING_INTERRUPTION" if the holder keeps
talking, or "INTERRUPTION" if the holder stops). "LAUGHTER" should
also be very short (e.g. "haha", "haha right").
- "timing_hint": exactly one of ["after_pause","latched","long_pause",
"mid_utterance","overlap_onset","during_holder"] — a QUALITATIVE
placement cue, never a number:
  - TURN utterances use "after_pause" (normal gap), "latched" (right
after, ~no gap), or "long_pause".
  - CLAIM_FLOOR uses "mid_utterance" or "overlap_onset".
  - VOCALIZE_NO_CLAIM uses "during_holder".
```

Stage-1 system prompt, verbatim (2/2): rules and exemplar

```
Hard rules:
1. The FIRST utterance must be a TURN (someone starts speaking).
2. A CLAIM_FLOOR or VOCALIZE_NO_CLAIM can only occur while someone
holds the floor (never first, and its "target" must be the current
holder — the most recent TURN speaker).
3. After a successful CLAIM_FLOOR (INTERRUPTION), the interrupter
normally continues with one or more TURN utterances (they now hold the
floor).
4. After a failed CLAIM_FLOOR (NON_FLOOR_TAKING_INTERRUPTION), the
ORIGINAL holder normally continues with a TURN (often finishing their
interrupted sentence, starting with "—").
5. EVERY listed speaker must speak at least one TURN.
6. Do NOT emit any tag other than the three above (no HOLDER_YIELDS
etc.) — clean turn hand-offs are implicit (a TURN by a new speaker
after a TURN means the previous speaker yielded the floor).
7. BACKCHANNELS must be <=3 words (ideally 1). A listener saying a
full phrase/sentence is a CLAIM_FLOOR, never a BACKCHANNEL.
8. Output ONLY a JSON object: {"turns": [ ... ]}. No prose before or
after.

Here is a short example of the exact output shape (a 3-speaker snippet
exercising every tag):
{"turns": [
 {"speaker": "speaker_1", "text": "Okay so for the mitosis question,
the key thing is the chromosome number stays the same.", "event_type":
"TURN", "timing_hint": "after_pause"},
 {"speaker": "speaker_2", "text": "Well hang on—", "event_type":
"CLAIM_FLOOR", "resolved_hint": "NON_FLOOR_TAKING_INTERRUPTION",
"target": "speaker_1", "timing_hint": "mid_utterance"},
 {"speaker": "speaker_1", "text": "—right, whereas meiosis halves it,
that's the whole point.", "event_type": "TURN", "timing_hint":
"latched"},
 {"speaker": "speaker_3", "text": "mm-hm", "event_type":
"VOCALIZE_NO_CLAIM", "sublabel": "BACKCHANNEL", "target": "speaker_1",
"timing_hint": "during_holder"},
 {"speaker": "speaker_2", "text": "Actually wait, I think you've got
prophase backwards, can we go back?", "event_type": "CLAIM_FLOOR",
"resolved_hint": "INTERRUPTION", "target": "speaker_1", "timing_hint":
"overlap_onset"},
 {"speaker": "speaker_2", "text": "In prophase one the homologs pair
up, that's what makes it different.", "event_type": "TURN",
"timing_hint": "latched"},
 {"speaker": "speaker_3", "text": "haha right", "event_type":
"VOCALIZE_NO_CLAIM", "sublabel": "LAUGHTER", "target": "speaker_2",
"timing_hint": "during_holder"},
 {"speaker": "speaker_1", "text": "Fair enough, good catch. So then
the diagram on slide four is wrong too.", "event_type": "TURN",
"timing_hint": "after_pause"}
]}
```

Figure 7: **Stage-1 system prompt**, shared by all 240 conversations. The LLM proposes symbolic acts only; it never assigns timestamps or final floor outcomes.

the balanced design, enabling evaluation across conversational styles without conflating them with speaker count.

Floor distribution. Adding speakers spreads the floor rather than speeding up exchanges. The share of floor-holding time taken by the most active speaker falls from 60.4% at $N{=}2$ to 51.4%, 41.0% and 36.4% at $N{=}3, 4, 5$, while speaker changes per minute stay between 7.6 and 9.1 (Table 9). Hold negatives become rarer with N (74 at $N{=}2$ vs. 48 at $N{=}5$) as mean floor segments shorten (7.07 to 6.26 s); mid-turn pauses are only inserted into segments of at least 4 s (Table 17). Listener activity grows over the same range (3.12 to 4.06 backchannels/min).

E Experimental Details

E.1 Offline Baseline Configuration

The offline suite covers twelve systems in three families of floor-control systems. Acoustic detectors include RMS energy, WebRTC VAD (Google WebRTC Project), and Silero VAD (Silero Team, 2024). Semantic endpoint detectors include Smart Turn v2 (Pipecat AI, 2025), Kyutai Semantic-VAD (Kyutai), and the Mimi Endpointer (Udupa et al., 2025). Turn-taking models include Voice Activity Projection (VAP) (Ekstedt and Skantze, 2022), MuVAP (Qi and Skantze, 2026), DualTurn (Rajaa, 2026), the WavLM-Large anchor (Chen et al., 2022), Turn-1-Mini (p99lab, 2026), and the ESPnet turn-taking model (Arora et al., 2025b). Table 22 gives the exact checkpoint, input configuration and score mapping of every system.

For multi-party evaluation, two-channel models

Stage-1 user prompt template

```
Write a {conversation_type} multi-party
conversation.

Scenario / domain: {domain}
Conversation style: {style string, Table 15}
Overall tone: {tone}
Number of speakers: {N}
Speakers (id: gender, speaking style — give each a
distinct personality/voice):
  - speaker_1: {gender}, {style}
  - . . .

Approximate soft targets (aim NEAR these; they are
guardrails, not exact quotas):
- Total utterances (TURN count): about {n_turns}
floor-holding turns over ~{D} minutes (mean turn
~{turn_len}s of speech, so ~{max(12,
2.5·turn_len)}-{4·turn_len} words each).
- Interruptions (CLAIM_FLOOR): about
{n_interruptions} across the conversation (mix of
INTERRUPTION and NON_FLOOR_TAKING_INTERRUPTION).
- Backchannels/laughter (VOCALIZE_NO_CLAIM): about
{n_backchannels}.

Make it feel like a real, specific conversation
about the scenario — concrete details, natural
disfluency, distinct personalities. Every speaker
participates. Output ONLY the JSON object {"turns":
[...]}.
```

Instantiated user prompt (test_00013)

```
Write a narrative multi-party conversation.
Scenario / domain: book/movie recommendation swap
Conversation style: storytelling with personal
anecdotes; longer turns, listeners backchanneling,
rare interruptions
Overall tone: professional and business-like
Number of speakers: 5
. . .
- Total utterances (TURN count): about 15
floor-holding turns over ~2.59 minutes . . .
- Interruptions (CLAIM_FLOOR): about 5 . . .
- Backchannels/laughter (VOCALIZE_NO_CLAIM): about
16.
```

Repair message (on validation failure)

```
That output had these problems;
fix them and return the full
corrected JSON:
- {error 1}
- . . . (up to 12 errors)
```

Figure 8: **Stage-1 user prompt.** The template is filled from the conversation specification and Table 15; the middle box shows the instantiated prompt of the narrative excerpt in Figure 6. On validation failure the repair message is appended and the model is queried again.

Component	Model	Realized (240 conv.)
Ambient bed	Procedural pink noise, scene-specific low-pass (200–2200 Hz), mains hum (100/200 Hz) for office scenes, slow ($<$0.2 Hz) drift; 10 scenes	SNR $\mathcal{U}(20, 38)$ dB: range 20.2–37.8, median 29.7
Discrete events	One random MUSAN sound-bible clip ($\leq$4 s) per sampled event slot, 12 dB below active speech, random time; $k \sim \{0, 1, 1, 2\}$ slots	0 / 1 / 2 slots: 49 / 124 / 67 conv.
Distant speech	2–4 faded MUSAN speech snippets of 1.5–3 s at 26 dB below active speech; $p = 0.2$	51 conversations

Table 19: **Stage-4 noise model.** SNR is measured against the active-speech RMS pooled over channels. Event-slot labels: keyboard clatter, notification chime, dishes, off-mic laughter, phone ring, door knock, chairs, school bell, alarm beep, paper shuffle (23–30 occurrences each). Scenes: outdoor park 33, living room TV 32, school hallway 27, office HVAC 25, studio room tone 24, cafe 22, office hum 22, dorm 20, quiet office 19, quiet apartment 16.

(VAP, MuVAP, Mimi Endpointer, DualTurn) operate in a speaker-versus-rest configuration, treating each participant as the candidate speaker and the mean of the remaining channels as the conversational context. Single-channel models run independently on every speaker channel; ESPnet is reported on the mono mixture of all channels. Systems are evaluated on the noisy, channel-separated MPTB recordings, and all inference is causal: each score is time-stamped at the end of the audio it has consumed. Except for VAP, which gates on its own voice-activity head, every learned INT score is gated by "another speaker is active" from WebRTC VAD (aggressiveness 2) on the other channels. Turn-1-Mini releases only its end-of-turn head and is evaluated only on EOT; no INT output is inferred from its predictions. All tunable detectors use thresholds selected on the 48-conversation development split and held fixed on the 192-conversation test split, following Section 4. Thresholds are not adjusted for individual speaker counts or interaction types. The threshold grids, selected thresholds and development scores are listed in Table 27.

E.2 Real-Audio Evaluation

To examine the transferability of results obtained from synthetic speech, we evaluate compatible offline detectors on two real-audio datasets: a dyadic subset of TurnBench (Jiang et al., 2026) comprising 12 conversations, and a four-party subset of AMI (Carletta et al., 2006) comprising 15 meetings (approximately 7.1 hours). The TurnBench subset uses consensus annotations, while AMI floor events are derived from its manual segment and word annotations using a documented floor-state mapping.

The same event definitions and scoring procedure are applied across datasets, with dataset-specific operating points selected where development data are available. Since the datasets differ in recording conditions, interaction structure, and speaker composition, cross-corpus comparisons characterize domain sensitivity rather than isolating

	Corpus Size		Interaction Dynamics					Benchmark Events	
Condition	**Dialogues**	**Duration (h)**	**Mean Turn (s)**	**Events/min**	**Backchannels/min**	**Interruptions/min**	**Overlap/dialogue (s)**	**EOT Pos./Neg.**	**INT Pos./Neg.**
By speaker count									
$N = 2$	60	1.90	7.07	13.8	3.12	0.70	10.2	853/74	80/356
$N = 3$	60	1.66	6.38	16.3	3.61	0.96	10.8	838/64	96/359
$N = 4$	60	1.67	6.10	16.3	3.60	0.94	10.9	875/55	94/361
$N = 5$	60	1.66	6.26	16.6	4.06	0.97	11.0	856/48	97/405
By interaction type									
Argumentative	40	1.11	6.29	15.2	2.95	1.08	10.7	520/49	72/196
Casual	40	1.10	4.81	19.4	4.24	0.92	11.3	739/37	61/280
Collaborative	40	1.07	5.32	17.8	3.43	1.33	10.6	610/37	85/220
Instructional	40	1.17	7.79	13.8	3.81	0.70	11.3	458/45	49/268
Narrative	40	1.41	9.22	11.8	3.34	0.53	11.5	471/45	45/282
Task-Oriented	40	1.03	5.27	17.5	3.80	0.89	9.1	624/28	55/235
All	**240**	**6.89**	**6.45**	**15.7**	**3.58**	**0.89**	**10.7**	**3422/241**	**367/1481**

Table 20: **Corpus composition and turn-taking dynamics across speaker counts and interaction types.** MPTB maintains balanced coverage across $N = 2$–5 while exhibiting relatively stable conversational activity as speaker count increases. Interaction types induce more distinct floor-management patterns: Casual conversations exhibit faster exchanges and shorter turns, Collaborative conversations have the highest interruption rate, and Narrative conversations sustain the longest turns with the fewest interruptions. EOT and INT $+/-$ indicate positive anchors and hard-negative spans, respectively.

Quantity (s)	n	**p10**	**p50**	**p90**
Floor-holding segment	4,030	2.19	5.18	10.65
Yield gap (next onset − offset)	3,182	−0.09	0.01	0.48
Hold pause span (EOT−)	241	0.49	0.83	1.12
Interruption overlap (onset→cut-off)	367	0.43	0.74	1.09
Failed claim duration	605	0.80	1.63	3.72
Backchannel duration (INT−)	1,176	0.44	0.64	1.04
Laughter duration (INT−)	305	0.53	0.97	1.82

Table 21: **Realized timing distributions** over all 240 conversations. Conversations last 48.2–208.2 s (median 99.9 s). Negative yield gaps are `latched` hand-overs that begin before the previous holder finishes.

a causal effect of speaker count.

TurnBench slice. Consensus anchors are rebuilt from the three released annotator tracks with ±200 ms endpoint agreement, following TurnBench's protocol. The slice contains 287 EOT positives, 571 hold spans, 107 INT positives and 1,272 negatives.

AMI floor-state mapping. Each AMI headset channel is one speaker. A single floor holder is tracked over the manual transcriber segments with the thresholds in Table 23; the resulting events pass through the same anchor derivation and scorer as MPTB. Over the 15 meetings this yields 1,458 EOT positives, 862 hold spans, 1,295 INT positives and 1,255 INT negatives (17.3 events, 2.95 backchannels and 3.04 interruptions per minute).

E.3 Full-Duplex Behavioral Evaluation

Scenario Design. We construct 120 interactive scenarios spanning four event conditions: true end-of-turn, mid-turn hold pause, successful interruption, and backchannel (30 each). Scenarios are balanced across $N \in \{2, 3, 4\}$ and the six interaction types. Each scenario is evaluated three times on five full-duplex systems: GPT-Live-1 (OpenAI, 2026), Gemini-3.8-Live (Google, 2026), MiniCPM-o-4.5 (Cui et al., 2026), Realtime-Venus (Zhao et al., 2026a), and Step-Audio-3-Realtime (Lin et al., 2026a), yielding 1,800 attempted trials. Agents receive streaming speech without access to scenario labels or reference event times.

START and STOP Probes. The START probe measures whether an agent correctly enters the conversational floor. A response following a genuine EOT is counted as a correct start, while speaking during a mid-turn hold pause constitutes a false start. We denote the corresponding rates by S and FS.

The STOP probe measures whether an agent relinquishes the floor appropriately. The agent is first induced to produce a multi-second response, during which an incoming speaker either successfully interrupts or produces a backchannel. Yielding to a genuine interruption constitutes a correct stop; yielding to a backchannel constitutes a false stop. We denote these rates by Y and FY.

Both probes use the balanced-accuracy and action-bias definitions in Section 4, with $(\text{Recall}, \text{FPR}) = (S, FS)$ for START and (Y, FY) for STOP.

System	Checkpoint / implementation	Input and context	EOT score	INT score
RMS Energy	frame RMS (dBFS)	per channel, 20 ms frames	speech→silence offset after 0.3 s silent hangover	speech onset while another channel is voiced, confirmed after 0.1 s
WebRTC VAD	`py-webrtcvad`	per channel, 16 kHz, 20 ms frames; θ = aggressiveness	as RMS, on WebRTC masks	as RMS, on WebRTC masks
Silero VAD	Silero VAD	per channel, 16 kHz; θ = speech prob.	as RMS, on Silero masks	as RMS, on Silero masks
Smart Turn v2	`pipecat-ai/smart-turn-v2` (wav2vec2-base, attention pooling)	per channel, 16 kHz, $\leq$8 s window ending at each energy-VAD offset ($-$40 dBFS, 0.3 s hangover)	$P(\text{complete})$	$1 - P(\text{complete})$ in TurnBench's Smart Turn pipeline (Silero VAD, 512-sample chunks)
Kyutai SVAD	`kyutai/stt-1b-en_fr-candle`	per channel, streaming Mimi, 12.5 Hz	$P(\text{end of turn})$ (VAD head 2)	$1 - P(\text{end of turn})$
Mimi Endpointer	`viks66/mimi-endpointer` (`oto_d1f`), two-stream LSTM over Mimi	s vs. mean of rest, 24 kHz, 25 Hz	$1 - P(\text{user})$	$P(\text{user})$
VAP	TurnBench fine-tuned VAP (`viks66/VAP_checkpoints`, `oto`)	s vs. mean of rest, 16 kHz stereo, 50 Hz; 20 s context, 5 s step	$1 - p_{\text{now}}[s]$	$p_{\text{now}}[s]$, gated by VAP $\text{vad}[\text{rest}] > 0.5$
MuVAP	`Haotian-Qi/MuVAP` `vap-speaker-mimi` (audio-only release)	s vs. mean of rest, 16 kHz, 25 Hz; 20 s context, 5 s step	$1 - p_{\text{now}}[s]$ (0–0.6 s)	$p_{\text{now}}[s]$
DualTurn	`anyreach-ai/dualturn-qwen2.5-mimi-0.5B`	s vs. mean of rest, 24 kHz, 12.5 Hz	`eot_probs`$[s]$	`hold_probs`$[s]$ (mapping chosen on dev)
WavLM Anchor	`ZhuoyanTao/causal-wavlm-turn-taking` (frozen WavLM-Large, AR decoder)	per channel, 16 kHz, 4 s window, 40 ms stride	$P(\text{NA}) + P(\text{T})$	$1 - P(\text{C})$
Turn-1-Mini	`p99lab/turn-1-mini` (released end-of-turn head)	per channel, 16 kHz, 4 kHz low-pass, 160 ms streaming steps	$P(\text{end of turn})$	not released
ESPnet	`espnet/Turn_taking_prediction_SWBD` (Whisper-medium encoder (Radford et al., 2023), 5-class head)	mono mix of all channels, 16 kHz, $\leq$30 s window, 40 ms hop	$P(\text{T}) \times$ energy share of s	$P(\text{I}) \times$ onset of $s \times$ others' activity

Table 22: **Offline system configurations.** Classes C/NA/I/BC/T = continuation / no-activity / interruption / backchannel / turn-shift. Every score is converted to decisions by rising edges above θ with a 2 s refractory period. Mappings follow TurnBench's reference adapters wherever one exists.

Rule	Threshold
Same-speaker segments merged (contiguous)	gap < 0.15 s
Mid-turn pause $\rightarrow$ TURN–hold (EOT$-$)	gap ≥ 0.20 s
Overlap counted as "during" the holder	≥ 0.10 s
Backchannel (INT$-$): duration / words	≤ 1.20 s / ≤ 3
Interruption (INT$+$): interrupter speaks at least	1.00 s
and holder falls silent within	2.50 s of onset
Other overlapping listener speech	non-floor-taking (excluded)
Non-holder starts after holder silence	holder yields (EOT$+$)

Table 23: **AMI floor-state mapping** from manual segments to MPTB events.

Instrumentation and Trial Validity. Provider-specific adapters record incoming audio timestamps, assistant speech intervals, emitted audio, and available interruption or cancellation events on a common monotonic clock. START trials require an observable opportunity for the agent to respond; STOP trials additionally require the agent to be speaking when the probe is injected. Trials failing the relevant observability conditions are excluded from behavioral scoring.

F Additional Results

F.1 Synthetic vs Real-world Turn Taking in Multi-party Setting

Table 24 places MPTB next to three real-speech sources. *TurnBench official* reports the published TurnBench leaderboard (real dyadic studio speech; 116 test conversations; TurnBench's own runs and operating points). *TurnBench dev slice* applies our scorer and our detector implementations to 12 real dyadic development conversations with consensus anchors (287 EOT positives / 571 hold spans; 107 INT positives / 1,272 negatives). *AMI meetings* uses 15 four-party headset meetings (7.09 h; ES2002a–d, ES2003a, ES2004a, ES2005a, IS1000a–b, IS1001a–b, TS3003a–b, TS3004a–b) with floor events derived from the manual segment annotations (1,458 EOT positives / 862 hold spans; 1,295 INT positives / 1,255 negatives; mapping in Appendix E.2). On both real slices thresholds are tuned in-slice.

The single system evaluated on all four sources is VAP. Its EOT recall decreases from 92.6% (MPTB) to 84.5% (TurnBench official), 79.1% (dev slice) and 52.1% (AMI). Its INT recall is 94.5% on official dyadic TurnBench, 61.7% on the dev slice, 35.9% on four-party AMI and 11.7% on MPTB, with FPR between 2.8% and 10.7% in every case. AMI is also denser in interruptions than MPTB (3.04 vs. 0.89 per minute).

	MPTB (ours)		TurnBench official		TurnBench dev slice		AMI meetings	
Audio	synthetic TTS + noise		real, studio		real, studio		real, headset	
Speakers	$N = 2$–5		$N = 2$		$N = 2$		$N = 4$	
Labels	generator script (exact)		3-annotator consensus		3-annotator consensus		derived from manual segments	
Threshold	MPTB dev → test		TB dev → test		tuned in-slice		tuned in-slice	
Conversations	192 test		116 test		12		15	
System	**EOT**	**INT**	**EOT**	**INT**	**EOT**	**INT**	**EOT**	**INT**
RMS Energy	49.8 / 40.9	66.2 / 25.0	71.8 / 63.2	99.6 / 44.5	—	—	28.9 / 17.4	30.0 / 8.2
Smart Turn	82.5 / 11.8	67.6 / 64.3	75.2 / 4.7	10.7 / 9.3	28.6 / 7.2	—	27.3 / 4.2	—
Kyutai SVAD	92.5 / 4.3	56.2 / 6.6	77.3 / 5.9	89.8 / 8.1	86.1 / 31.3	—	62.2 / 41.1	—
VAP	92.6 / 7.5	11.7 / 2.8	84.5 / 5.5	94.5 / 10.7	79.1 / 26.4	61.7 / 5.1	52.1 / 8.2	35.9 / 5.3
Mimi Endpointer	90.5 / 3.8	13.9 / 3.8	78.2 / 7.8	89.9 / 10.6	—	—	—	—
WavLM Anchor	90.4 / 22.6	12.8 / 7.3	80.0 / 5.4	86.8 / 5.4	—	—	—	—
ESPnet Turn-Taking	86.9 / 12.4	1.1 / 1.9	82.6 / 7.8	57.3 / 8.0	—	—	—	—

Table 24: **Interruption detection remains challenging in natural multi-party speech.** Recall / FPR (%) for systems evaluated on synthetic MPTB, the official TurnBench leaderboard (real dyadic speech; TurnBench's own runs), our TurnBench dev slice, and real four-party AMI meetings. Gray : within that source's FPR budget. Sources differ in audio, labels and operating-point selection (header rows), and both real slices tune thresholds on the evaluated data. Absolute numbers are therefore not comparable across columns but only in directions. Every learned system has lower EOT recall on real audio than on MPTB. Interruptions, however, collapse in multi-party speech on both real and synthetic audio. VAP, the one learned system evaluated on every source, detects 94.5% of interruptions in real dyadic TurnBench, but only 35.9% in real four-party AMI, even with in-slice tuning, and 11.7% on MPTB. VAP, Mimi Endpointer and WavLM Anchor are trained on TurnBench data and are in-domain there.

F.2 EOT and INT detection recall-FPR breakdown by interaction type

Figure 9 reports recall and FPR for all twelve systems in each interaction type, and Figure 10 isolates the five systems that are within budget on both tracks. Each cell uses the pooled operating point; per-type cells contain 32 test conversations, 364–601 EOT positives, 21–38 hold spans, 32–66 INT positives and 154–225 INT negatives.

F.3 Performance by party count

Table 25 reports VAP by party count. Table 26 extends the analysis to every system. Each N contributes 48 test conversations (EOT: 695/661/676/684 positives and 55/48/44/39 hold spans; INT: 60/73/72/76 positives and 276/288/277/331 negatives for $N = 2..5$). Three patterns stand out. (i) Kyutai's EOT recall rises from 84.6% to 96.6% with FPR at or below 9.1% throughout, while its INT recall falls from 63.3% to 48.7% together with its INT FPR (10.5% to 3.9%). (ii) MuVAP's INT recall gain from 23.3% to 40.8% is accompanied by an FPR increase from 5.4% to 12.1%; DualTurn gains 6.2 points with FPR between 6.9% and 10.5%. (iii) Acoustic VADs recover more interruptions as N grows (WebRTC 70.0% to 93.4%, Silero 73.3% to 92.1%) at a constant 82–90% INT FPR, since more speakers produce more overlap onsets of every kind.

N	EOT recall/FPR	INT recall/FPR
2	.93/.04	.07/.03
3	.93/.04	.10/.04
4	.92/.16	.11/.03
5	.92/.08	.18/.02

Table 25: VAP test performance by party count. The N=4 EOT FPR exceeds the nominal budget after the dev-selected threshold is frozen.

F.4 Operating Points

Table 27 lists the frozen threshold of every system together with development and test recall/FPR. Two learned systems satisfy the development constraint on EOT but exceed the test ceiling: WavLM Anchor (dev FPR 7.3%, test 22.6%) and Turn-1-Mini (dev 9.1%, test 25.3%). Smart Turn's INT inversion has no threshold with dev FPR below 58.9%.

F.5 Latency Distributions

Table 28 reports the full signed-latency distribution of matched detections. The positive-anchor window admits detections from -250 to $+3000$ ms, so p10 values below zero indicate anticipatory firing.

(a) EOT recall (%) ↑

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy	56.2	47.3	45.6	54.7	44.1	52.6
WebRTC VAD	90.6	90.5	92.1	92.6	91.7	93.5
Silero VAD	97.0	96.8	97.1	97.3	97.3	97.8
Smart Turn v2	81.3	80.4	83.2	83.8	83.4	83.8
Kyutai SVAD	95.1	91.0	90.5	94.0	96.0	90.6
Mimi Endpointer	90.4	92.2	89.2	88.7	90.4	91.2
VAP	92.9	93.7	92.5	91.5	91.2	93.0
MuVAP	86.2	88.9	89.4	87.4	88.0	88.3
DualTurn	89.2	90.0	89.2	87.4	89.6	90.6
WavLM Anchor	91.1	91.2	88.8	89.6	91.2	90.4
Turn-1-Mini	89.2	84.9	86.3	86.0	89.0	87.1
ESPnet Turn-Taking	87.4	89.4	86.5	84.1	82.9	88.8

(b) INT recall (%) ↑

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy	61.8	71.1	65.2	75.0	53.1	69.8
WebRTC VAD	89.1	91.1	78.8	85.0	75.0	86.0
Silero VAD	89.1	93.3	83.3	82.5	75.0	86.0
Smart Turn v2	67.3	73.3	56.1	70.0	84.4	65.1
Kyutai SVAD	54.5	53.3	47.0	77.5	59.4	53.5
Mimi Endpointer	9.1	15.6	15.2	15.0	15.6	14.0
VAP	7.3	15.6	9.1	15.0	9.4	16.3
MuVAP	30.9	37.8	34.8	35.0	25.0	41.9
DualTurn	36.4	40.0	33.3	47.5	34.4	32.6
WavLM Anchor	12.7	6.7	21.2	7.5	15.6	9.3
Turn-1-Mini	—	—	—	—	—	—
ESPnet Turn-Taking	0.0	0.0	1.5	2.5	0.0	2.3

(c) EOT false-positive rate (%) ↓

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy	34.2	25.0	37.0	45.7	51.4	52.4
WebRTC VAD	86.8	82.1	92.6	97.1	91.9	90.5
Silero VAD	97.4	96.4	92.6	94.3	89.2	90.5
Smart Turn v2	13.2	17.9	18.5	8.6	5.4	9.5
Kyutai SVAD	7.9	3.6	7.4	2.9	2.7	0.0
Mimi Endpointer	5.3	0.0	7.4	0.0	8.1	0.0
VAP	13.2	3.6	3.7	8.6	10.8	0.0
MuVAP	18.4	0.0	18.5	2.9	10.8	9.5
DualTurn	15.8	7.1	14.8	2.9	8.1	19.0
WavLM Anchor	26.3	28.6	14.8	25.7	13.5	28.6
Turn-1-Mini	34.2	14.3	29.6	22.9	27.0	19.0
ESPnet Turn-Taking	15.8	10.7	11.1	11.4	16.2	4.8

(d) INT false-positive rate (%) ↓

	Argum.	Casual	Collab.	Instr.	Narr.	Task
RMS Energy	25.3	27.1	19.9	30.0	24.5	21.7
WebRTC VAD	83.1	87.1	80.1	86.6	81.5	83.2
Silero VAD	87.7	91.1	90.3	89.9	88.0	88.0
Smart Turn v2	66.9	62.7	61.4	64.1	69.0	62.0
Kyutai SVAD	4.5	7.6	7.4	6.0	7.4	6.0
Mimi Endpointer	2.6	8.0	3.4	2.3	3.2	2.7
VAP	0.0	4.4	0.6	3.2	5.1	2.2
MuVAP	5.2	9.3	8.5	11.5	11.1	4.3
DualTurn	3.2	10.2	8.5	10.1	8.8	6.5
WavLM Anchor	9.7	5.8	6.3	6.5	9.3	6.5
Turn-1-Mini	—	—	—	—	—	—
ESPnet Turn-Taking	3.2	2.7	2.8	0.0	0.9	2.2

Figure 9: **EOT and INT performance across interaction types.** Recall (blue) and false-positive rate (orange) per conversation type on the test split (32 conversations per type), at each system's pooled operating point. Darker means higher; scales are shared across panels. EOT recall barely depends on the regime: every learned system stays within 6.5 points across types (VAP 91.2–93.7%). INT does depend on it. The two strongest INT systems both peak on asymmetric *instructional* talk (Kyutai 77.5%, DualTurn 47.5%), and Kyutai drops to 47.0% in overlap-heavy *collaborative* talk. Learned INT false alarms stay at or below 11.5% in every type (excluding the Smart Turn inversion). For Kyutai, Mimi-EP and DualTurn they peak in backchannel-dense *casual* conversation, as TurnBench reports for dyadic speech. Per-type EOT FPR rests on only 21–38 negative spans, so differences between types there are not reliable. Kyutai and Smart Turn INT are inversions of end-of-turn scores; — marks no INT output.

Smart Turn scores candidate turn ends proposed by an energy VAD and stamps each decision at the start of the silence, before the VAD's 0.3 s confirmation hangover; its EOT latencies therefore exclude that hangover and can be negative. Among in-budget systems, VAP is the only one with a sub-200 ms median EOT decision; Kyutai and Mimi Endpointer commit over a second after the turn ends.

F.6 Paired Comparisons

Table 29 reports paired conversation-cluster bootstrap differences against VAP, each system at its own operating point. On EOT, Kyutai and VAP are statistically indistinguishable in recall (-0.1 points, CI $[-1.5, 1.3]$), while every other learned system has significantly lower recall than VAP. On INT, Kyutai (+44.5), DualTurn (+25.3) and MuVAP (+22.8) significantly exceed VAP, each with a small but significant FPR increase (3.8–5.8 points); Mimi Endpointer and WavLM Anchor do not differ from VAP in INT recall.

F.7 Robustness to Acoustic Conditions

Table 30 slices the five systems in budget on both tracks, and Silero VAD as an acoustic reference, by ambient SNR and by the presence of distant third-party speech. Within the rendered 20–38 dB range, no learned system shows a monotonic loss of EOT recall at lower SNR (VAP: 92.4–92.9% in every bin). Distant speech raises learned INT FPR by at most 1.2 points (Mimi Endpointer 3.6% to

System	EOT recall / FPR (%)				INT recall / FPR (%)			
	$N{=}2$	$N{=}3$	$N{=}4$	$N{=}5$	$N{=}2$	$N{=}3$	$N{=}4$	$N{=}5$
Acoustic VAD								
RMS Energy	48.5 / 38.2	61.1 / 62.5	41.9 / 25.0	48.1 / 35.9	55.0 / 22.8	74.0 / 34.4	66.7 / 18.8	67.1 / 23.9
WebRTC VAD	92.1 / 85.5	93.8 / 95.8	90.4 / 86.4	90.9 / 94.9	70.0 / 83.0	82.2 / 83.0	88.9 / 82.7	93.4 / 86.1
Silero VAD	97.8 / 94.5	98.0 / 93.8	96.0 / 95.5	96.9 / 89.7	73.3 / 89.1	82.2 / 89.2	91.7 / 88.4	92.1 / 90.0
Semantic endpointers								
Smart Turn v2	83.5 / 18.2	82.3 / 10.4	83.3 / 11.4	81.0 / 5.1	55.0 / 59.4	67.1 / 60.4	68.1 / 71.1	77.6 / 66.2
Kyutai SVAD	84.6 / 1.8	92.1 / 4.2	96.9 / 9.1	96.6 / 2.6	63.3 / 10.5	63.0 / 6.6	51.4 / 5.8	48.7 / 3.9
Mimi Endpointer	90.2 / 3.6	89.7 / 4.2	92.0 / 2.3	90.1 / 5.1	13.3 / 5.8	19.2 / 2.1	12.5 / 4.7	10.5 / 3.0
Turn-taking models								
VAP	93.2 / 3.6	93.2 / 4.2	92.0 / 15.9	92.0 / 7.7	6.7 / 2.5	9.6 / 3.8	11.1 / 3.2	18.4 / 1.8
MuVAP	88.6 / 5.5	86.8 / 8.3	88.8 / 13.6	88.3 / 15.4	23.3 / 5.4	32.9 / 7.6	38.9 / 8.7	40.8 / 12.1
DualTurn	89.4 / 7.3	89.4 / 14.6	89.3 / 9.1	89.6 / 12.8	33.3 / 8.0	35.6 / 6.9	38.9 / 10.5	39.5 / 7.6
WavLM Anchor	90.2 / 29.1	90.5 / 22.9	90.1 / 15.9	90.8 / 20.5	11.7 / 5.8	11.0 / 6.2	11.1 / 7.2	17.1 / 9.4
Turn-1-Mini	86.3 / 20.0	85.6 / 31.2	89.6 / 22.7	86.0 / 28.2	—	—	—	—
ESPnet Turn-Taking (mix)	86.6 / 7.3	85.6 / 18.8	89.2 / 13.6	86.0 / 10.3	1.7 / 1.1	1.4 / 3.8	0.0 / 1.8	1.3 / 0.9
ESPnet Turn-Taking (per-ch.)	4.0 / 9.1	5.4 / 2.1	4.6 / 6.8	4.8 / 7.7	5.0 / 6.2	6.8 / 7.3	4.2 / 6.1	3.9 / 4.5

Table 26: **Recall / FPR by party count for every system** (test split, pooled operating point). The ESPnet per-channel variant runs the same checkpoint on each isolated channel instead of the mono mix.

System	EOT			INT		
	θ	Dev R / F	Test R / F	θ	Dev R / F	Test R / F
RMS Energy	-50 dBFS	44.9 / 30.9	49.8 / 40.9	-50 dBFS	51.2 / 19.7	66.2 / 25.0
WebRTC VAD	aggr. 3	91.6 / 87.3	91.8 / 90.3	aggr. 3	83.7 / 84.5	84.3 / 83.8
Silero VAD	0.30	97.6 / 90.9	97.2 / 93.5	0.70	83.7 / 86.4	85.4 / 89.2
Smart Turn v2	0.50	78.3 / 14.5	82.5 / 11.8	0.0442	73.3 / 58.9	67.6 / 64.3
Kyutai SVAD	0.95	93.2 / 3.6	92.5 / 4.3	0.9991	65.1 / 6.1	56.2 / 6.6
Mimi Endpointer	0.95	90.9 / 1.8	90.5 / 3.8	0.99	25.6 / 3.2	13.9 / 3.8
VAP	0.80	92.9 / 9.1	92.6 / 7.5	0.90	12.8 / 3.6	11.7 / 2.8
MuVAP	0.95	88.5 / 5.5	88.1 / 10.2	0.90	33.7 / 5.8	34.5 / 8.6
DualTurn	0.40	90.5 / 7.3	89.4 / 10.8	0.9711	45.3 / 6.8	37.0 / 8.2
WavLM Anchor	0.75	91.5 / 7.3	90.4 / 22.6	0.80	15.1 / 8.7	12.8 / 7.3
Turn-1-Mini	0.35	87.7 / 9.1	86.9 / 25.3	—	—	—
ESPnet Turn-Taking (mix)	0.00136	87.0 / 9.1	86.9 / 12.4	0.50	2.3 / 1.6	1.1 / 1.9
ESPnet Turn-Taking (per-ch.)	0.70	6.1 / 7.3	4.7 / 6.5	0.50	4.7 / 8.1	5.0 / 6.0

Table 27: **Dev-selected operating points.** R/F = recall/FPR (%). θ is the score threshold (dBFS for RMS, WebRTC aggressiveness level, probability otherwise). Gray rows are in budget on both tracks. Candidate grids: RMS $\{-50, -45, \ldots, -30\}$ dBFS; WebRTC $\{0, 1, 2, 3\}$; Silero $\{0.3, \ldots, 0.7\}$; Kyutai and Smart Turn EOT $\{0.3, \ldots, 0.9, 0.95\}$; VAP and MuVAP $\{0.5, \ldots, 0.9, 0.95\}$; other learned systems $\{0.05, 0.10, \ldots, 0.95, 0.97, 0.98, 0.99, 0.995\}$ plus dev-score quantiles.

4.8%); for VAP it is 2.8% with and without, and for Kyutai, MuVAP and DualTurn it is lower when distant speech is present. Floor-control errors on MPTB are therefore not explained by the acoustic nuisance layer.

F.8 Error Anatomy

To localize errors, we join every scored test anchor with its script utterance and score each attribute slice separately (Table 32).

End-of-turn scores detect sustained listener speech, not floor claims. Kyutai's INT score fires on 0.2% of one-word listener vocalizations, 5.7% of two- or three-word ones, and 72.3% of negatives with four or more words, all of which are laughter vocalizations such as "*Haha, yeah, I've been waiting all week for this one*". Its recall on true interruptions also grows with claim length (37.2% for 1–3 words vs. 62.6% for 9+ words). Its INT advantage over VAP (Table 29) therefore comes largely from duration, not from modelling floor ownership. DualTurn and MuVAP show the same direction with flatter slopes (26.2% FPR on 4+-word negatives), and all learned systems false-alarm more on laughter than on backchannels except ESPnet.

Hold pauses after complete questions look like yields. For Kyutai, Mimi Endpointer, MuVAP, DualTurn and Smart Turn, false EOT triggers on hold pauses are most frequent when the pre-pause clause ends in a question mark (33 spans; e.g., DualTurn 18.2% vs. 10.8% after a period, Smart Turn 24.2% vs. 11.7%), and every learned system detects true yields after questions more reliably than yields

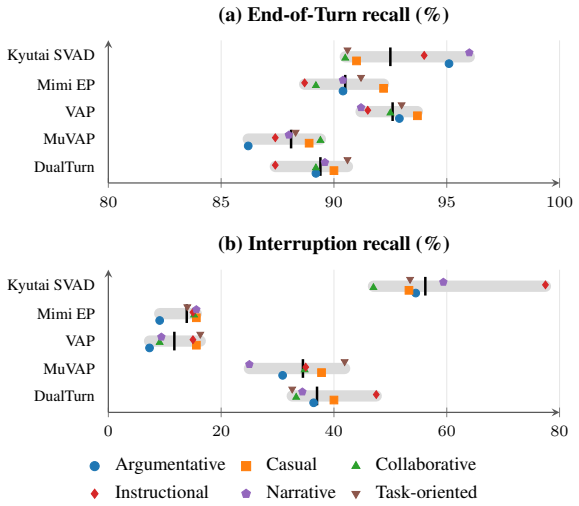


Figure 10: **Interaction type matters for interruptions, not for turn ends.** Recall per conversation type for the five systems within the FPR budget on both tracks (test split, 32 conversations per type). Gray bars span each system's worst-to-best type and black ticks mark pooled recall. EOT recall varies by at most 5.5 points across types. INT recall swings by up to 30.5 points (Kyutai: 47.0% collaborative vs. 77.5% instructional). The hardest INT type is model-specific: argumentative for VAP and Mimi EP, collaborative for Kyutai, narrative for MuVAP, and task-oriented for DualTurn.

after statements (e.g., VAP 96.9% vs. 90.8%). This is consistent with reliance on clause completion, a cue that a hold after a question does not follow. WavLM Anchor false-alarms on 20.5–23.4% of hold spans of every clause type and Turn-1-Mini on 27.3–29.7% after complete clauses, which is the source of their over-budget test FPR.

Contested holds are the multi-party EOT failure. In 28 of the 186 test hold spans, another listener's failed floor claim overlaps the holder's pause, so the floor is briefly contested before the holder resumes. These contested holds account for most false EOT triggers of the speaker-versus-rest models (Table 31): VAP false-alarms on 39.3% of them versus 1.9% of uncontested holds (11 of its 14 false alarms), MuVAP on 57.1% versus 1.9%, and DualTurn on 42.9% versus 5.1%. Kyutai, which scores each channel in isolation, is nearly unaffected (7.1% vs. 3.8%). These models read a competing voice in the pooled "rest" channel as the holder releasing the floor. Contested holds make up 10.9% of hold spans at $N{=}2$ and 15.4–18.2% at $N{=}3$–5, and with more listeners the competing voice can come from any of $N-1$ channels that the pooled rest stream merges.

System	EOT (ms) p10	EOT (ms) p50	EOT (ms) p90	INT (ms) p10	INT (ms) p50	INT (ms) p90
RMS Energy	267	292	1653	−38	859	2572
WebRTC VAD	224	332	380	114	128	146
Silero VAD	359	397	435	92	119	198
Smart Turn v2	−32	−18	614	55	87	446
Kyutai SVAD	666	1829	1960	432	689	1121
Mimi Endpointer	878	1231	1630	346	870	2298
VAP	−9	143	555	299	492	1340
MuVAP	90	553	1563	164	618	1129
DualTurn	404	777	1812	224	572	1122
WavLM Anchor	576	1185	1734	−167	42	971
Turn-1-Mini	635	1087	1384	—	—	—
ESPnet (mix)	482	1084	2205	66	71	88

Table 28: **Signed latency** $t_{\text{pred}} - t_{\text{gold}}$ of matched detections on the test split. ESPnet INT percentiles rest on 3 matched detections.

System − VAP		Δ Recall	Δ FPR
EOT	Kyutai SVAD	−0.1 [−1.5, 1.3]	−3.2 [−7.7, 1.1]
	Mimi EP	−2.1 [−2.9, −1.3]	−3.8 [−8.1, 0.5]
	DualTurn	−3.2 [−4.0, −2.3]	+3.2 [−1.6, 8.2]
	MuVAP	−4.5 [−5.6, −3.4]	+2.7 [−1.8, 7.2]
	WavLM	−2.2 [−3.0, −1.4]	+15.1 [7.0, 23.1]
	Turn-1-Mini	−5.7 [−6.9, −4.5]	+17.7 [10.2, 25.5]
	ESPnet (mix)	−5.7 [−7.0, −4.5]	+4.8 [−0.6, 10.2]
INT	Kyutai SVAD	+44.5 [37.5, 51.3]	+3.8 [2.0, 5.5]
	DualTurn	+25.3 [18.2, 32.2]	+5.4 [3.5, 7.3]
	MuVAP	+22.8 [17.8, 28.0]	+5.8 [4.2, 7.5]
	Mimi EP	+2.1 [−2.2, 6.6]	+1.0 [−0.3, 2.4]
	WavLM	+1.1 [−4.1, 6.1]	+4.4 [2.6, 6.3]
	ESPnet (mix)	−10.7 [−14.5, −7.1]	−0.9 [−2.3, 0.3]
	Smart Turn v2	+55.9 [49.1, 62.5]	+61.5 [58.5, 64.4]

Table 29: **Paired differences to VAP** (points, 95% CI; $B = 10{,}000$ conversation-cluster resamples of the 192 test conversations).

G Full-Duplex Behavioral Evaluation Details

This section documents the 1,800-trial behavioral study of Table 4 (protocol in Appendix E.3). Table 33 lists every protocol constant, and Figure 11 gives the exact instructions and spoken prompts presented to the agents.

Agent configuration. Gemini-3.8-Live is accessed through the Live WebSocket API with the provider's default voice activity handling (voice *Puck*). GPT-Live-1 uses OpenAI's Live session API (voice *marin*); it arbitrates the floor internally and exposes no cancel events. Step-Audio-3-Realtime uses the OpenAI-compatible realtime endpoint with server VAD (500 ms silence). MiniCPM-o-4.5 and Realtime-Venus run locally on one H100 in duplex mode. All agents receive the same audio. Speech intervals are measured from a virtual playout of the audio each agent emits, never from provider control events.

System	EOT recall by SNR (dB)				INT recall by SNR (dB)				EOT FPR, distant speech		INT FPR, distant speech	
	20–25	25–30	30–35	35–40	20–25	25–30	30–35	35–40	absent	present	absent	present
Kyutai SVAD	91.9	94.5	92.8	89.6	52.4	56.7	59.3	55.3	3.9	6.1	6.8	5.6
Mimi Endpointer	91.2	91.5	89.2	89.8	12.7	13.3	16.0	12.8	4.6	0.0	3.6	4.8
VAP	92.7	92.5	92.4	92.9	12.7	10.0	13.6	10.6	7.8	6.1	2.8	2.8
MuVAP	88.8	88.7	87.2	87.6	25.4	33.3	40.7	38.3	10.5	9.1	9.3	6.0
DualTurn	89.7	89.6	89.5	88.8	34.9	34.4	39.5	40.4	10.5	12.1	8.8	6.0
Silero VAD†	96.9	97.6	95.9	98.2	82.5	83.3	87.7	91.5	93.5	93.9	90.2	85.3
Conversations	44	59	52	37	44	59	52	37	151	41	151	41

Table 30: **Robustness slices** (test split, %, pooled operating points). Signal to Noise Ratio (SNR) bins refer to the ambient bed.

System	EOT FPR on holds (%) contested (n=28)	uncontested (n=158)	Share of false alarms
Kyutai SVAD	7.1	3.8	2/8
Mimi Endpointer	25.0	0.0	7/7
VAP	39.3	1.9	11/14
MuVAP	57.1	1.9	16/19
DualTurn	42.9	5.1	12/20
WavLM Anchor	17.9	23.4	5/42
Turn-1-Mini	32.1	24.1	9/47
ESPnet (mix)	35.7	8.2	10/23

Table 31: **Contested holds.** A hold is contested when another speaker's non-floor-taking claim overlaps the pause span. Last column: contested false alarms / all EOT false alarms on the test split.

G.1 Per-family counts and accuracy

Table 34 breaks accuracy down by party count and reports run-to-run consistency (the fraction of scenarios on which all valid repeats agree). The STOP failures are stable properties of each agent, not noise: MiniCPM-o ignores 73 of 74 valid interruptions with perfect consistency, and Gemini yields to 86 of 88 backchannels. GPT-Live-1 is the least deterministic agent on STOP (consistency .40 on interruptions and .60 on backchannels), and its yield rate grows with N (35% at N=2 to 65% at N=4) together with its backchannel accuracy (48% to 75%).

G.2 Latency

Table 35 reports START latency as the agent's speech onset relative to the end of the genuine turn, and STOP latency as the time from injection of the interrupting speech to the end of the agent's ongoing utterance, over all valid trials. Gemini stops 230 ms (median) after an interruption begins, in line with provider-side cancellation on detected input speech (Table 36). MiniCPM-o (median 8.6 s) and Venus (3.0 s) finish their ongoing answer before going silent, which is why their interruption accuracy is near zero under the 1.5 s stop window.

Session instruction (all agents, all trials)

You are taking part in a spoken conversation with other people. Participate naturally. Respond when another participant clearly gives you the floor. Do not interrupt speakers who are still holding the floor.

STOP elicitation questions (spoken, one per scenario)

1. Could you briefly describe what happens when water boils?
2. Can you give me a short description of what a rainbow is?
3. Please briefly explain what clouds are made of.
4. Could you tell me in a few sentences what a library is for?
5. Can you briefly describe what happens during sunrise?
6. Please give me a short explanation of why leaves change colour.
7. Could you briefly say what a bicycle is used for?
8. Can you describe in a couple of sentences what snow is?
9. Please briefly explain what a map shows.
10. Could you give a short description of what a garden is?

Figure 11: **Agent-facing text.** The instruction never mentions turn-taking evaluation, interruptions or measurement. Questions elicit a 4–8 s answer that requires no reasoning, so the response content cannot confound the stop decision; only speech onset and cessation are scored.

G.3 Barge-in instrumentation

Table 36 separates provider-driven cancellation from model-driven yielding. Every Gemini stop on a true interruption, and 95.5% of its stops on backchannels, coincide with a provider barge-in event; its STOP behavior is therefore the provider's voice-activity cancellation policy, which cannot distinguish a backchannel from a takeover. The other agents expose no cancel events, so their stops are model decisions. In 61.7% of its valid interruption trials, GPT-Live-1 starts a new utterance within 2 s of ending the interrupted one, so a yield is often a brief pause followed by reclaiming the floor.

System	EOT (pre-pause / pre-yield clause ending)					INT FPR on negatives					INT recall by claim length		
	FPR on holds			Recall on yields		by word count			by type		words		
	“.”	“?”	“—”	“.”	“?”	1	2–3	4+	BC	Laugh	1–3	4–8	9+
Kyutai SVAD	1.8	9.1	7.7	92.2	93.6	0.2	5.7	72.3	2.4	21.7	37.2	55.6	62.6
Mimi Endpointer	2.7	9.1	2.6	88.9	95.0	1.5	4.5	20.0	2.6	8.3	23.3	12.1	12.2
VAP	9.9	9.1	0.0	90.8	96.9	1.2	3.7	10.8	2.6	3.6	9.3	14.1	10.8
MuVAP	9.9	15.2	7.7	86.2	92.9	6.0	9.4	26.2	7.3	13.4	27.9	37.4	34.5
DualTurn	10.8	18.2	5.1	87.9	92.9	2.9	12.1	26.2	6.9	13.0	27.9	36.4	40.3
WavLM Anchor	23.4	21.2	20.5	88.4	95.7	7.0	7.0	10.8	6.5	9.9	14.0	14.1	11.5
Turn-1-Mini	29.7	27.3	10.3	85.8	90.2	—	—	—	—	—	—	—	—
ESPnet (mix)	15.3	9.1	7.7	85.0	91.6	1.8	2.0	1.5	2.2	0.8	0.0	2.0	0.7
Smart Turn v2†	11.7	24.2	10.3	82.9	84.7	59.7	68.1	76.9	63.0	69.2	79.1	67.7	64.0
n (test)	111	33	39	1,921	717	596	511	65	919	253	43	99	139

Table 32: **Error anatomy** (test split, %). Clause ending is the final character of the script utterance before the hold pause or yield (“—” marks a self-interrupted clause). Word counts are taken from the script text of the listener vocalization or the interrupting claim. †Smart Turn EOT from the re-run on released audio at the frozen threshold.

Constant	Value
Scenario families × size	4 × 30 (balanced N=2, 3, 4)
Max. scenarios per source conversation	2
Selection seed	20260929
Repeats per scenario and agent	3
Stream audio	mono mix of all participants’ noisy channels, 24 kHz; the agent joins as an additional party
START context before target turn onset	20 s
START open floor after true EOT	4.0 s silence
START hold stimulus	real audio through resume + 3.0 s
START correct window	$[t - 0.25,\ t + 3.0]$ s
STOP context before prompt	12 s real conversation
STOP prompt	1 of 10 trivial questions, Kokoro voice unused in the conversation
Wait for agent speech	14 s
Injection delay after agent onset	2.0 s
Observation after injection	8.0 s
STOP correct window (agent stops)	$[0,\ 1.5]$ s after injection
Agent speech gap merged as jitter	< 0.35 s
GPT-Live output energy gate	≈ -60 dBFS
Local models (MiniCPM-o, Venus)	16 kHz, 1.0 s causal chunks, stream-time clock

Table 33: **Full-duplex protocol constants.** All constants are identical across agents; no protocol parameter was adapted to any model.

H Offline–Online Association

For four models with both offline and online measurements, model-level rank association is weak (Spearman $\rho = -0.20$ for EOT→START; $\rho = 0.316$ for INT→STOP). Event-level STOP association is stronger: behavioral correctness is .584 when the offline detector is correct versus .422 when it is wrong, OR 1.91 [1.34, 2.73], $n = 506$. START shows no such association (.800 vs. .820; OR .89 [.53,1.50], $n = 720$). We do not interpret the pooled association because START and STOP have different base rates.

Offline detector pass. The offline measurements come from a separate pass on the same 120 scenarios in which each agent acts as a detector: for EOT it hears the isolated channel of the speaker whose turn ends, for INT an agent-free mix, both with a detector instruction, and its response onset is read out as the commit time and scored with the MPTB window. Table 37 gives the event-level contingency counts behind the odds ratios.

H.1 Sim2Real Gap

We evaluate compatible detectors on real dyadic TurnBench (Jiang et al., 2026) and four-party AMI recordings to examine cross-corpus generalization. Results are in Table 24 and discussed in Appendix F.1; datasets and the AMI floor-state mapping are specified in Appendix E.2.

I Qualitative Analysis

This section shows verbatim test-split events together with the decision of every offline system at its frozen operating point (Figures 12–13) and of every full-duplex agent in each repeat (Figure 14). In each box, highlighted lines are the scored event, tags give the generator’s label, ● marks a correct decision and × an error, and grey numbers give the decision time in ms relative to the event onset (anchor or span start). Examples were selected to illustrate the systematic patterns quantified in Appendix F.8; the counts in each *Reading* line refer to the whole test split.

J Reproducibility Checklist

The release will include scenario manifests and seeds, synchronized input audio, exact event logs, detector thresholds selected on dev, provider/-

Model	true EOT (START +)				hold pause (START −)				true INT (STOP +)				backchannel (STOP −)			
	N=2	3	4	Cons.	N=2	3	4	Cons.	N=2	3	4	Cons.	N=2	3	4	Cons.
Gemini-3.8-Live	.93	.80	.93	.77	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.07	.00	.00	.93
GPT-Live-1	.95	.90	.95	.93	.95	1.00	1.00	.97	.35	.50	.65	.40	.48	.60	.75	.60
MiniCPM-o-4.5	.97	.77	.87	.77	.87	1.00	.97	.87	.04	.00	.00	1.00	1.00	1.00	1.00	1.00
Realtime-Venus	.70	.87	.87	.60	.93	.90	.90	.77	.13	.04	.07	.77	.85	.90	.88	.67
Step-Audio-3	.00	.00	.00	1.00	1.00	1.00	1.00	1.00	.00	.20	.00	1.00	1.00	.80	.80	.50

Table 34: **Behavioral accuracy by party count and run-to-run consistency (Cons.).** Accuracy is over valid trials in each N (10 scenarios $\times$ up to 3 repeats). Step-Audio-3 STOP cells rest on 2–5 valid trials each.

Model	Family	p10	p50	p90	n
Gemini	true EOT	1141	2036	2258	80
Gemini	true INT	200	230	263	90
GPT-Live-1	true EOT	980	1313	1796	56
GPT-Live-1	true INT	1094	1491	2204	60
MiniCPM-o	true EOT	442	870	1223	78
MiniCPM-o	true INT	2000	8600	11000	74
Venus	true EOT	510	900	1535	73
Venus	true INT	1624	3000	6000	74
Step-Audio-3	true INT	-197	3367	11307	11

Table 35: Selected response latencies (ms) for valid trials with measurable responses.

Model	INT barge-in	BC barge-in	Resume $<$ 2s
Gemini	1.000	.955	.022
GPT-Live-1	.000	.000	.617
MiniCPM-o	.000	.000	.122
Venus	.000	.000	.243
Step-Audio-3	.000	.000	.091

Table 36: Fraction of STOP events associated with provider barge-in/cancel events and short resumption rate after true interruptions.

Probe	Detector correct		Detector wrong	
	beh. ok	beh. wrong	beh. ok	beh. wrong
START	487	122	91	20
STOP	160	114	98	134

Table 37: **Event-level contingency** between offline detector correctness and behavioral correctness on the same scenario (valid trials of the four agents with an offline pass).

model identifiers and decoding parameters, one JSONL record per attempted trial, invalid-trial reason codes, the monotonic timestamp conversion used by every adapter. Table 38 maps each component to the appendix that specifies it.

(a) One takeover, split verdicts · N=5 · Instructional

test_00036 · INT positive on S2 at 44.80 s

S5 [non-floor-taking] I was wondering about that too.
S1 [turn] Great catch. We added a 'skip for now' button, very visible, right below the survey. [. . .] Actually, we're seeing about thirty percent take the survey up front, which honestly is much higher than the old flow.
S3 [non-floor-taking] Oh wow, that's awesome!
▷ **S2** [interruption] *Do you think that's because of the new placement, or is it just the nudge text?*
S2 [turn] Like, I'm curious if there's any evidence people feel more—uh, invested, I guess, with that up front?

Kyutai ● 717 MuVAP ● 277 DualTurn ● 1117 Smart Turn ● 605
VAP × Mimi EP × WavLM × ESPnet ×

Reading. The question follows a failed claim by S3. VAP and Mimi EP (INT FPR 2.8% and 3.8%) stay silent while Kyutai, MuVAP and DualTurn fire. At $N \geq 4$, 8 test interruptions show exactly this split.

(b) The easy dyadic case · N=2 · Narrative

test_00101 · INT positive on S2 at 20.93 s

S1 [turn] —uh, one student just blurted out, 'But how do we know the rainforest even matters to us here?' [. . .] honestly, it was the liveliest they'd ever been.
S2 [backchannel] That's fantastic.
▷ **S2** [interruption] *Did you let them just run with it?*
S2 [turn] Because honestly, those are the moments when you see them making connections, right?

Kyutai ● 266 VAP ● 306 MuVAP ● 306 DualTurn ● 347
Mimi EP ● 387
Smart Turn ● 795 WavLM × ESPnet × RMS/WebRTC/Silero ×

Reading. With one possible claimant, every in-budget two-channel model fires within 0.4 s. The acoustic VADs miss it: S2's backchannel ends 80 ms before the claim, so no separate speech onset exists. Such cases are rare: VAP detects 6.7% of N=2 interruptions overall.

(c) Invisible to every learned model · N=5 · Casual

test_00031 · INT positive on S5 at 18.79 s

S4 [turn] Go ahead, we'd love that.
S3 [backchannel] Yeah, please do.
S1 [turn] Okay, cool. Uh, I'm Ben, I just wrapped up three years at Pinecreek—they're a little startup in logistics? [. . .] I really like figuring out how teams tick, you know?
▷ **S5** [interruption] *Oh man, Pinecreek's that one with, like, the weird office dog, right?*
S5 [turn] I think I saw them on TikTok with a basset hound in a conference room. That's you guys, right?
S1 [turn] Yeah! Chip. He snores through every all-hands.

Kyutai × Mimi EP × VAP × MuVAP × DualTurn ×
WavLM × ESPnet × Smart Turn × RMS ● 128 WebRTC ● 128
Silero ● 187

Reading. A topical, enthusiastic takeover from a fifth participant who had not spoken yet. 23 of the 281 test interruptions (8.2%) are missed by every learned system with an INT output; here only the acoustic overlap-onset rules (INT FPR 25.0–89.2%) fire.

(d) Laughter that sounds like a claim · N=5 · Instructional

test_00036 · INT negative span on S3, 62.13–65.26 s

S1 [turn] Yeah, and that's the cool part. [. . .] it feels more like, 'help us help you,' not just a wall of questions.
S5 [backchannel] That makes sense.
▷ **S3** [laughter] *Haha, yeah, walls of questions scare me off every time.*
S1 [turn] Totally. So, after preferences, they hit the guided profile setup [. . .]

VAP ● WavLM ● ESPnet ● Kyutai × 2270 Mimi EP × 2229
MuVAP × 1990 DualTurn × 2669 Smart Turn × 77
RMS/WebRTC/Silero ×

Reading. A three-second laughing comment never takes the floor, but four learned systems fire after about two seconds of it. Kyutai fires on 72.3% of such 4+-word laughter spans and on 0.2% of one-word listener vocalizations.

Figure 12: **Interruption track: qualitative cases.** (a)–(c) positives, (d) a hard negative. Decisions use each system's frozen test operating point.

Component	Released artifact / setting	Specified in
Scenario sampling	240 specifications (N, type, domain, tone, duration, voices, noise profile, seed); 10 per (N, type) cell	App. C.3, Tab. 16
Script generation	`gpt-4.1`, temperature 0.9, 8,000 tokens, ≤3 attempts; prompts verbatim; validated scripts released	Figs. 7, 8, Tab. 15
Orchestration	All timing distributions; per-conversation seed; render sidecar with every clip placement	Tab. 17
Speech synthesis	Kokoro-82M, 24 kHz, speed 1.0, voice binding, content-hashed clip cache	Tab. 18
Noise rendering	Scene, SNR, event slots, distant-speech placements and MUSAN clip ids per conversation; noise seeds	Tab. 19
Gold and anchors	Event logs; deterministic anchor derivation and scorer	App. B.1, B.2
Splits	48 dev / 192 test, voice-disjoint	Tab. 10
Offline systems	Checkpoint ids, input configuration, score mappings, threshold grids, frozen θ	Tab. 22, 27
Statistics	Conversation-cluster bootstrap, B = 10,000, seeds 20261010 (CIs), 20261011–20261015 and 20261111 (paired tests)	App. B.2, F.6
Full-duplex study	120 scenario packs (stream, inject, metadata), selection seed 20260929, protocol constants, instruction and prompts, adapters, one record per trial with invalid reasons; bootstrap seed 20261012	App. G
Real-audio slices	TurnBench consensus reconstruction (±200 ms); AMI meeting list and floor-state thresholds	App. E.2
Compute	NVIDIA H100 80GB GPUs for TTS, learned detectors and local full-duplex agents; API agents via provider endpoints	—

Table 38: **Reproducibility map.** All non-generative stages are deterministic given the released seeds.

(a) Stacked backchannels · $N{=}5$ · Narrative

test_00013 · INT negative span on S2, 142.81–143.75 s

S5 [turn] Okay, so, uh, on a totally different note, if you're interested in documentaries, I'd suggest *American Factory*. [. . .]
S1 [turn] Yeah, I saw that last year. It was sobering, but so well done.
▷ **S2** [backchannel] *Definitely.*
S3 [backchannel] right
S4 [backchannel] Uh-huh.
S2 [turn] It sounds like most of us go for the heavy stuff—maybe next time someone should bring a comedy, just for balance.

Kyutai ● Mimi EP ● DualTurn ● ESPnet ●
VAP × 387 MuVAP × 427 WavLM × 187 Smart Turn × 611
RMS/WebRTC/Silero ×

Reading. Three listeners acknowledge S1 within 1.7 s (142.81, 143.43, 144.46 s), a stack that needs at least three listeners. S2 does take the next turn after S1 yields, so a projection model can be right about the next speaker and still fire on the wrong event.

(b) A contested hold · $N{=}5$ · Narrative

test_00038 · EOT hold span on S4, 83.89–84.95 s

S5 [turn] —that's a good thought, yeah. I think it could feel more like a conversation then, not a demand.
S3 [backchannel] mm
▷ **S4** [turn, hold] *So do we want to try a version where it's there but, like, ambient? Just an icon in the summary, and maybe if you haven't used it in a while, it gets a little nudge?*
S1 [non-floor-taking] That sounds reasonable to me, but maybe we need a quick user test—
S4 [turn] —or perhaps we just put that in this week's sprint as an experiment?

Kyutai ● Smart Turn ● ESPnet ● VAP × 188 MuVAP × 268
DualTurn × 428 Turn-1-Mini × 588 WavLM × 828 Mimi EP × 948
RMS/WebRTC/Silero ×

Reading. S4 ends on a question and S1 starts answering (81.87–85.59 s) during the pause, yet S4 keeps the floor. Six learned systems call a turn end. Contested holds are 15% of test hold spans but produce 11 of VAP's 14 and 16 of MuVAP's 19 false EOT alarms.

(c) An unfinished clause · $N{=}2$ · Narrative

test_00005 · EOT hold span on S1, 42.14–42.68 s

S2 [turn] But that's exactly the kind of random detail these trivia nights love.
S1 [backchannel] Exactly
▷ **S1** [turn, hold] *Actually, the one that always gets me is the sports round. I just blank out. Like, who scored the most points in—*
S2 [non-floor-taking] —LeBron, right?
S1 [turn] —NBA history, yeah, but somehow I always write Michael Jordan out of habit!

All 9 learned systems ● RMS × 259 WebRTC × 359 Silero × 420

Reading. The same silence that fools every acoustic VAD is rejected by every learned model, because the clause is syntactically open. VAP never false-alarms on a hold after a self-interrupted clause (0 of 39).

Figure 13: **Hard negatives: qualitative cases.** (a) INT negative; (b)–(c) EOT holds. Learned endpointers reject incomplete clauses but confuse a contested pause with a yield.

(a) STOP − · a one-word backchannel

fdb_backchannel_098 · $N=2$ · Argumentative

Context [S1] But isn't silence its own statement? I mean, how am I supposed to feel supported if he just vanishes [. . .]?

Agent [speaking] (answering the trivial question)

▷ **Inject** [S2, backchannel] *right*

Correct = keep speaking.
Gemini ××× 220–240
GPT-Live-1 ●●●
MiniCPM-o ●●●
Venus ●●●
Step-Audio-3 — no valid trial

Reading. Gemini stops within 240 ms in every repeat, each time with a provider barge-in event. Over the study it yields to 86 of 88 backchannels, 84 of them via barge-in.

(b) STOP + · a genuine takeover

fdb_true_int_086 · $N=4$ · Casual

Context [S4] Haha, yes! The bagel. Such a trip.

Agent [speaking] (answering the trivial question)

▷ **Inject** [S1, interruption] *You're selling it, seriously. I need to—*

Correct = stop within 1.5 s.
Gemini ●●● 199–220
GPT-Live-1 ●● 1000–1221
MiniCPM-o ××× stops at 9.8–11.0 s
Venus ××× stops at 2.7–5.0 s
Step-Audio-3 — no valid trial

Reading. MiniCPM-o and Venus talk through the takeover and finish their answer; MiniCPM-o does so on 73 of 74 valid interruptions.

(c) Detector vs. participant

Step-Audio-3-Realtime · all 120 scenarios

Offline EOT (detector)	29/30 ●
Offline hold rejection	30/30 ●
START: speaks after true EOT	0/90 ×
START: silent in hold	90/90 ●
STOP: valid trials	24/180

Reading. Step-Audio-3 recognizes turn ends when asked to detect them, but as a participant it never takes the floor within 3 s and rarely answers the STOP question within 14 s. Its START bACC of 50.0% reflects inaction, not discrimination (bias −1.0).

(d) Same prompt, different run

GPT-Live-1 · STOP + consistency

On 60% of interruption scenarios GPT-Live-1's two valid repeats disagree (e.g., fdb_true_int_061: ×●). Its yield rate rises from 35% at $N=2$ to 65% at $N=4$.

Figure 14: **Full-duplex agents: qualitative cases.** Each mark is one repeat of the same scenario. Grey numbers are the time from injection to the end of the agent's ongoing utterance (ms unless stated).